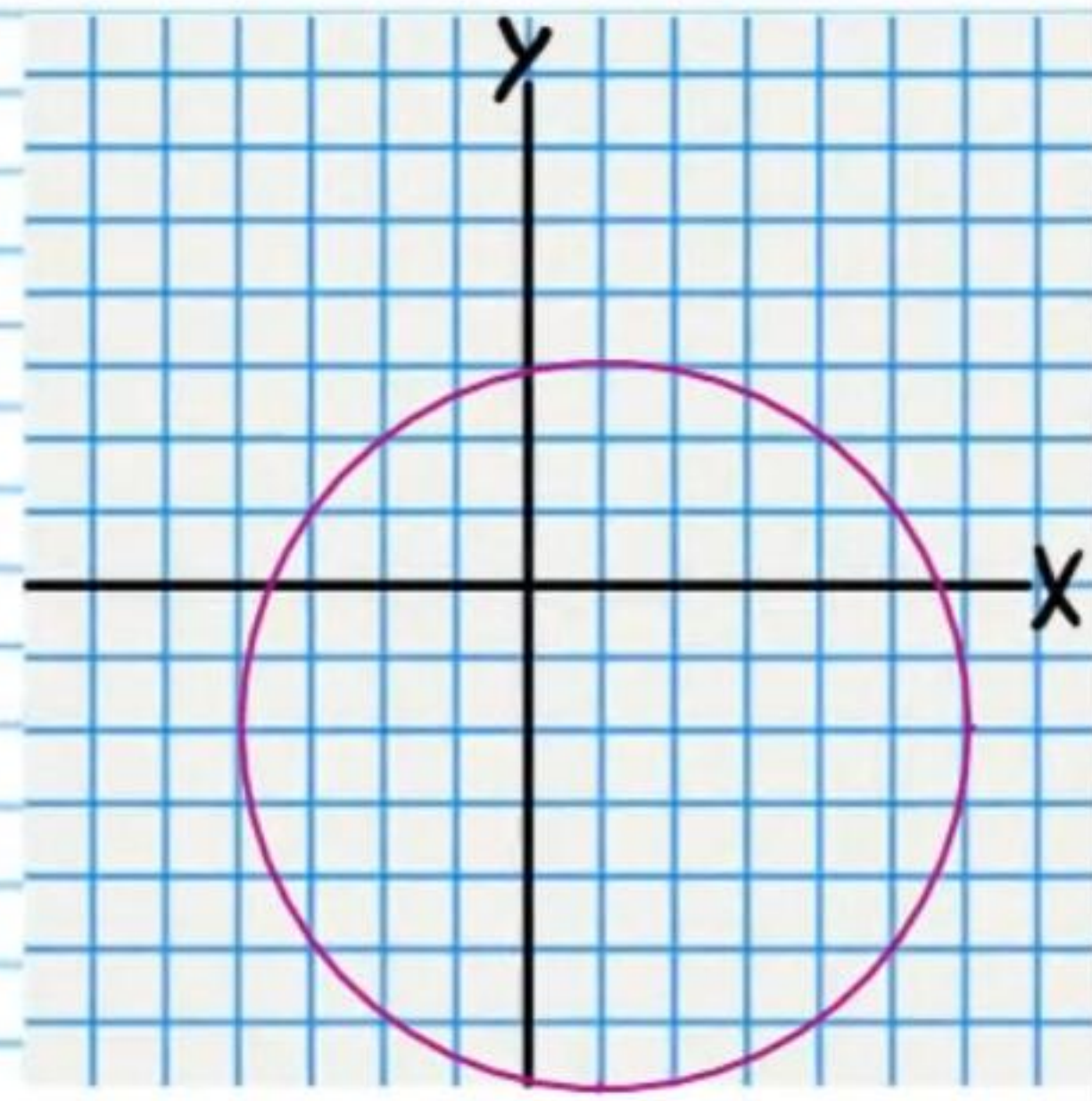


The Conics (Part II) Homework

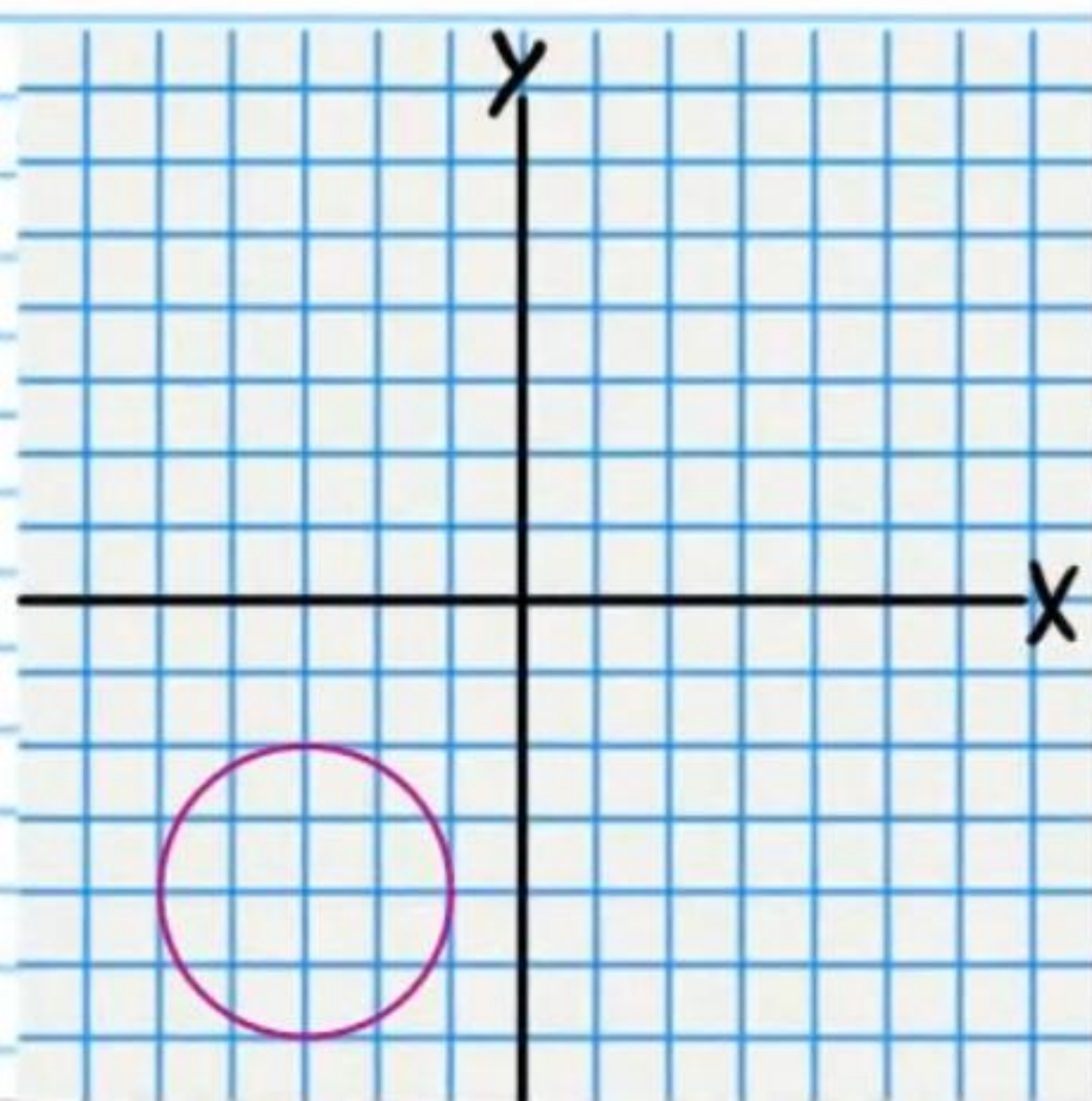
Name
Date
Course

Write an equation of each graph.

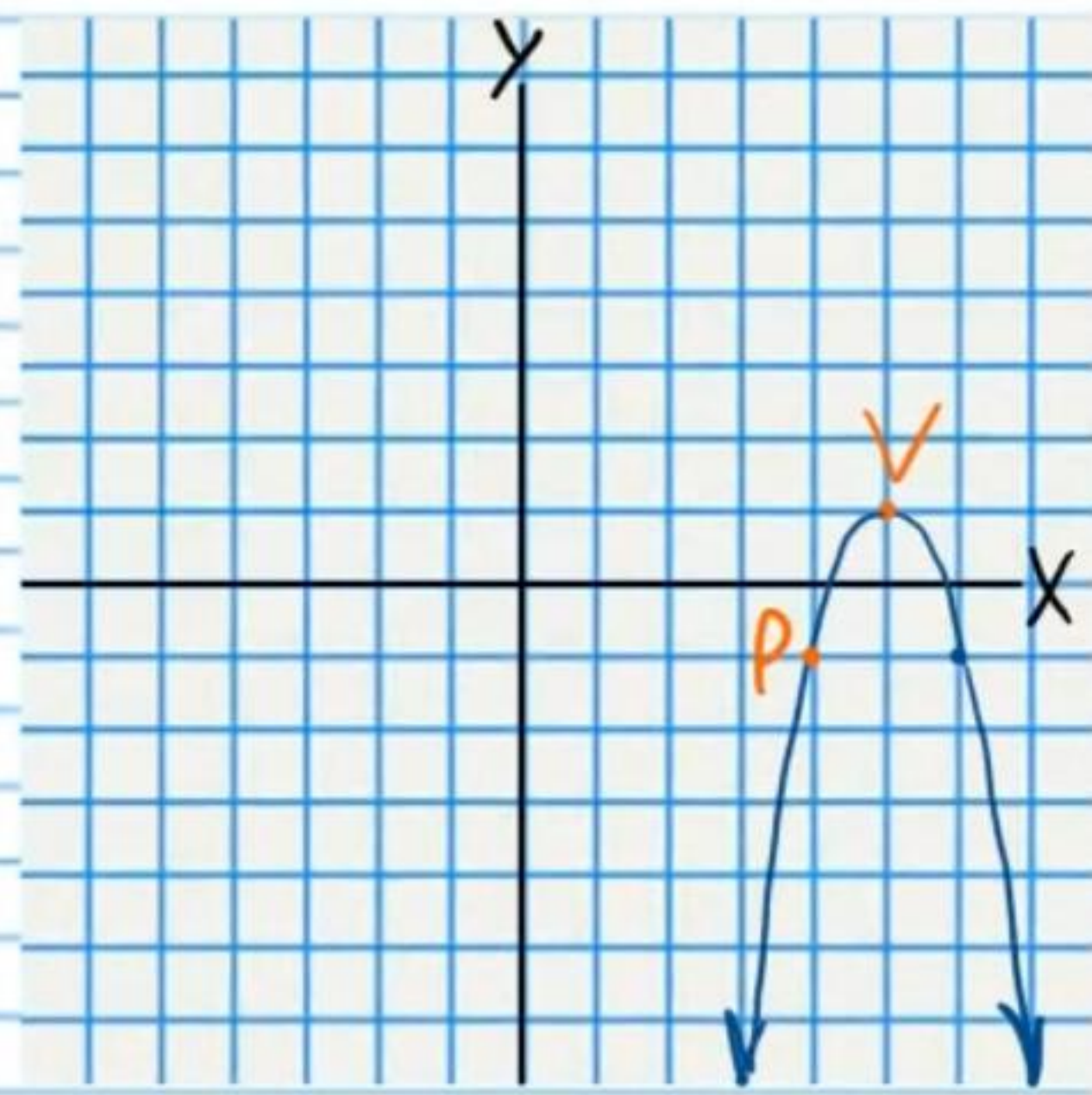
①



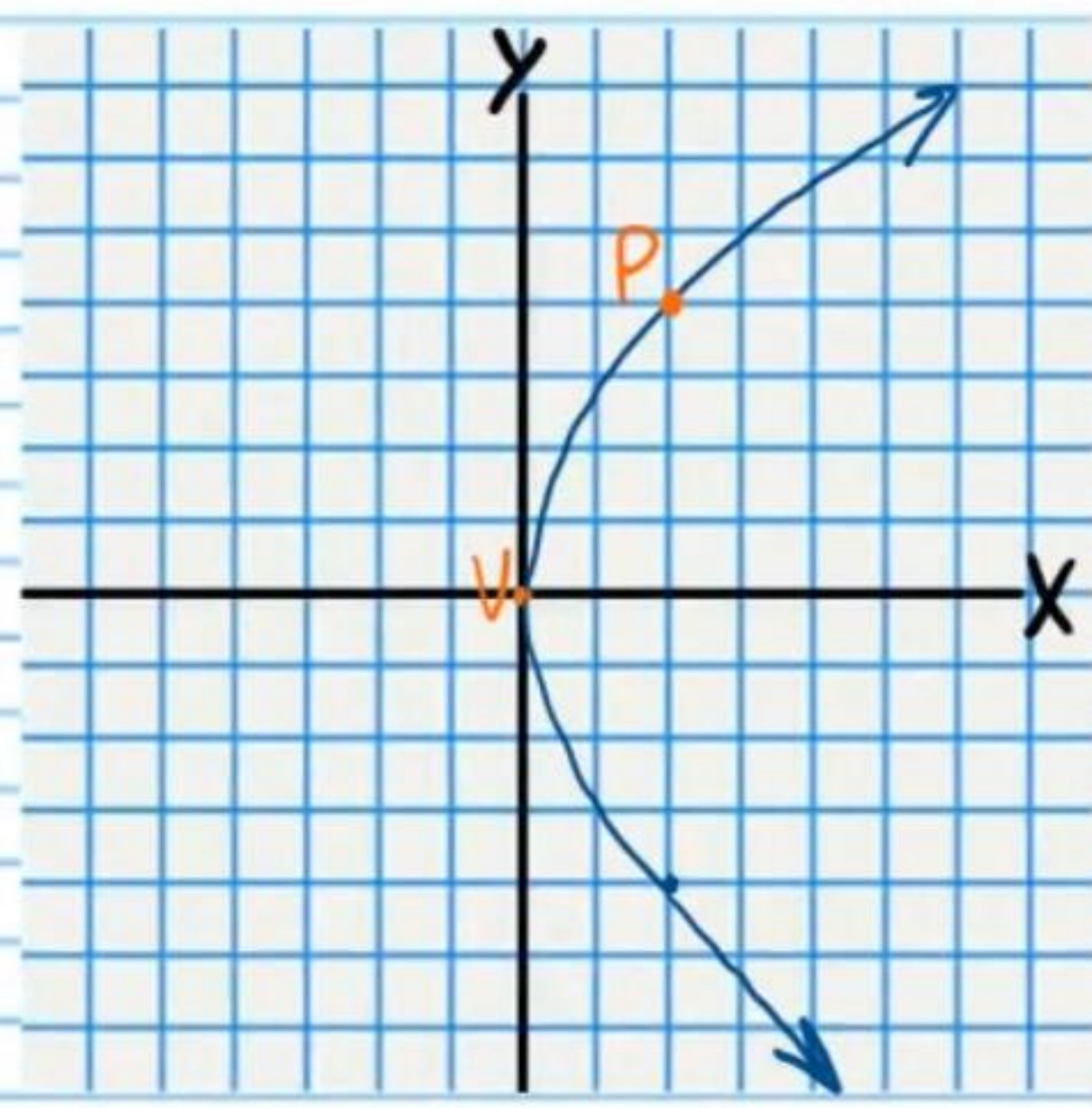
②



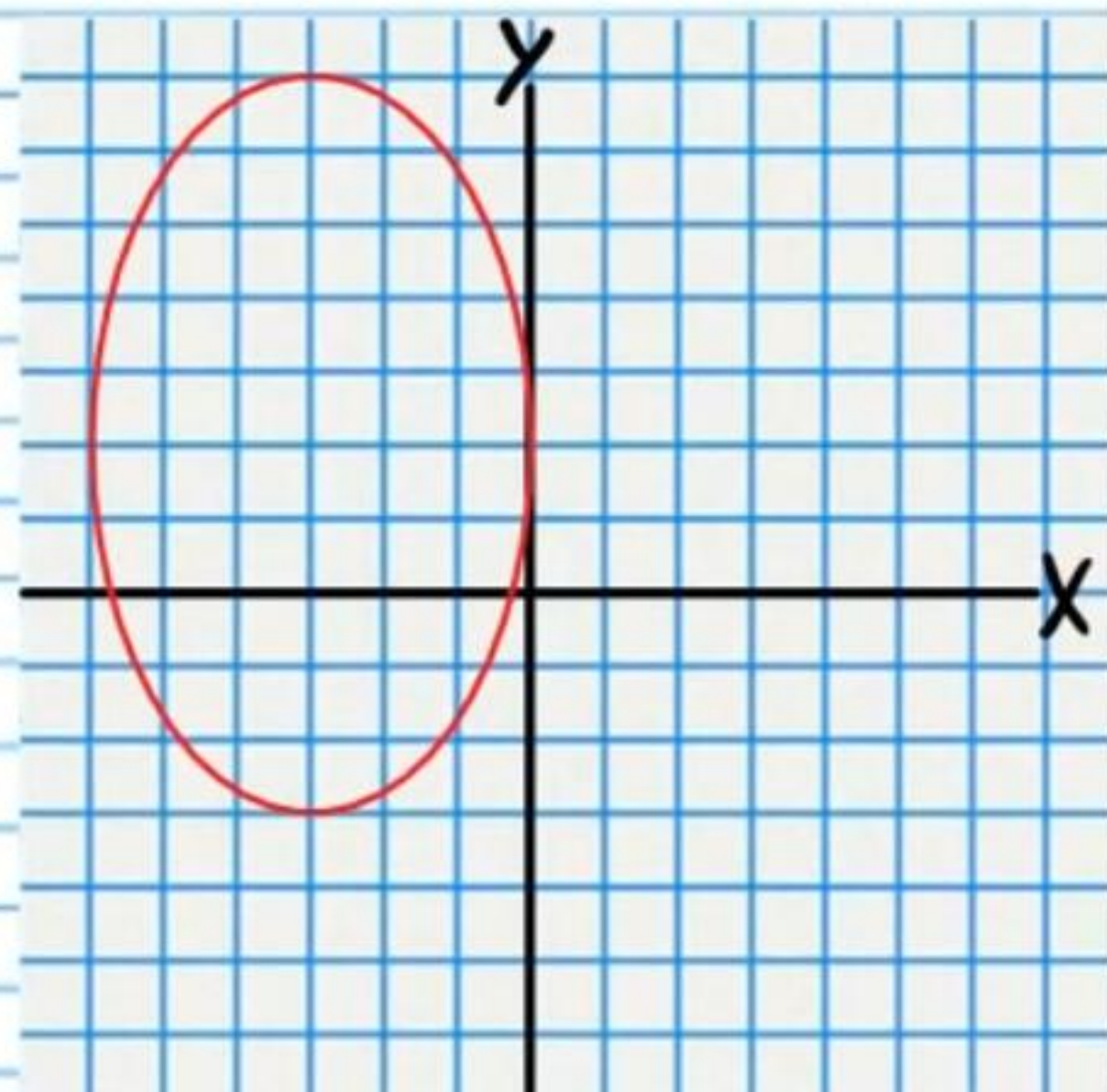
③



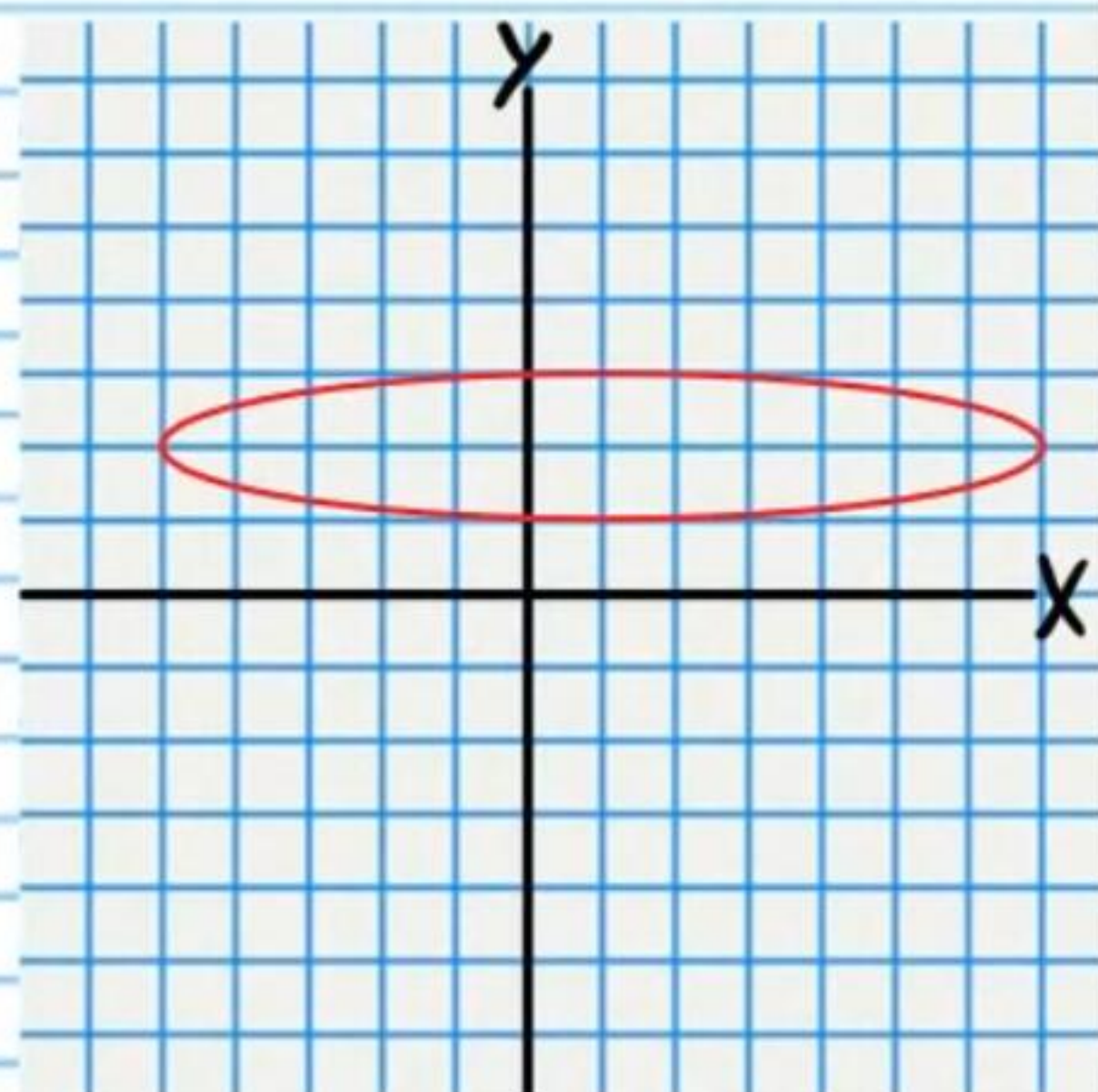
④



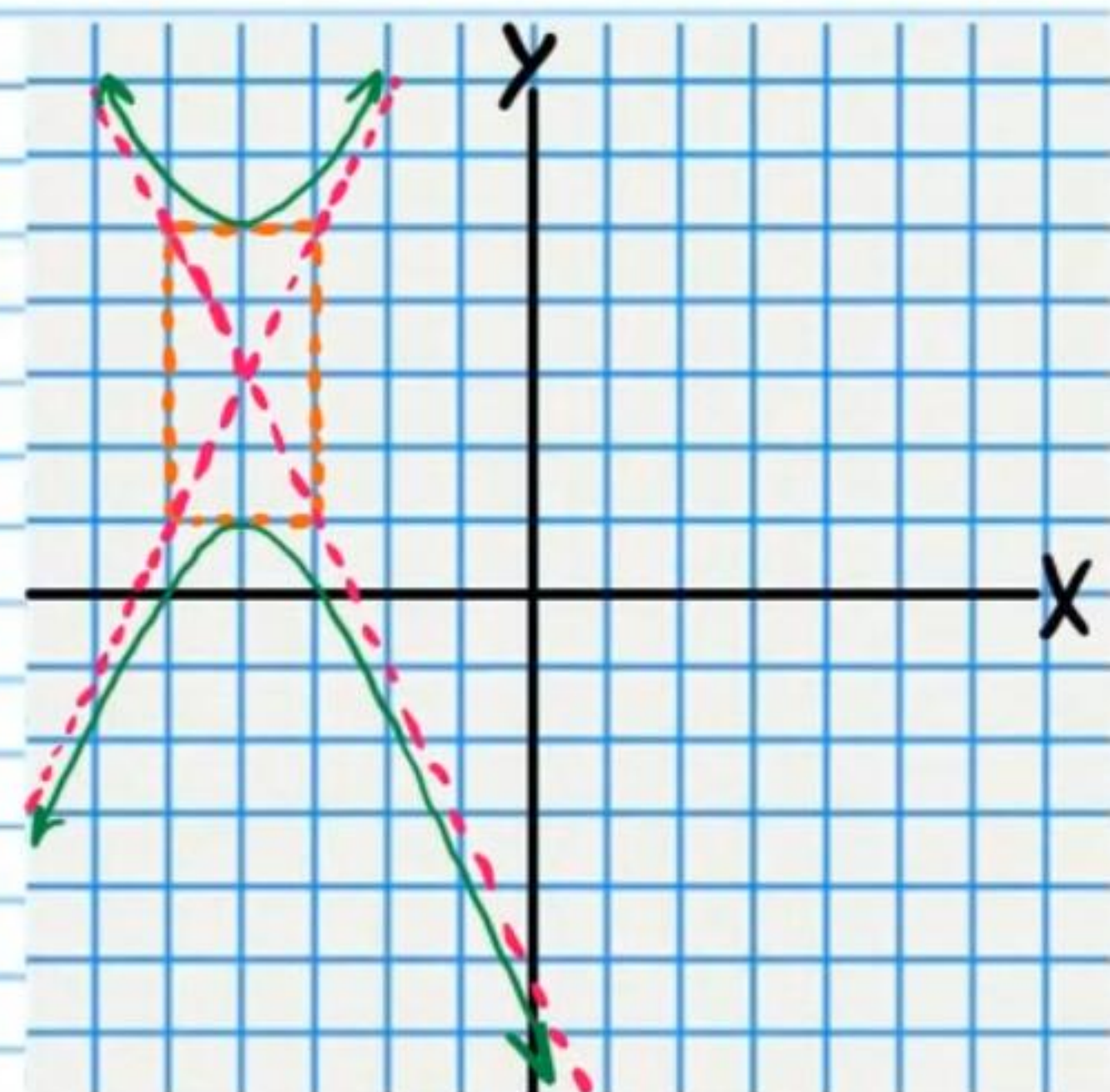
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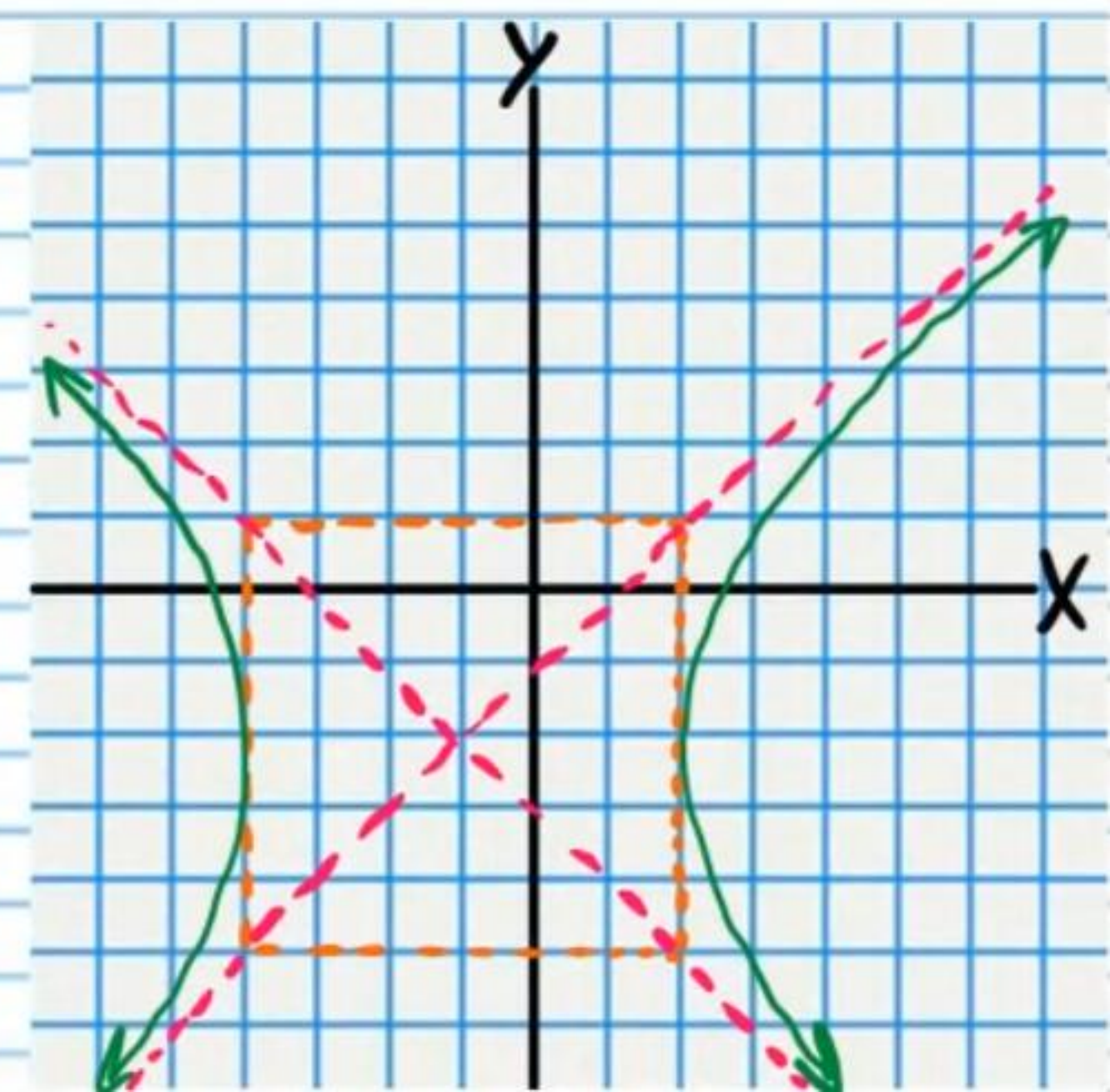
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7

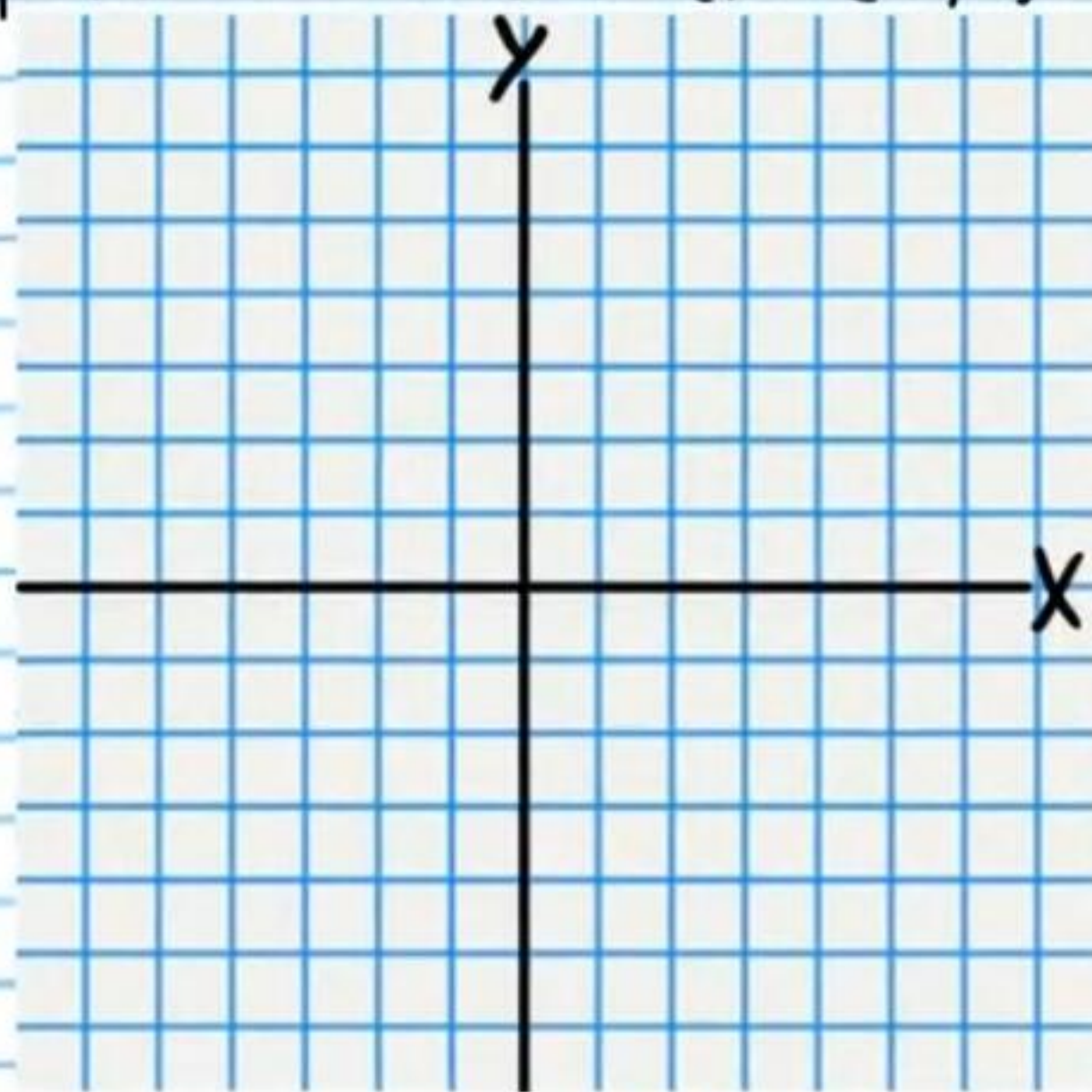


8

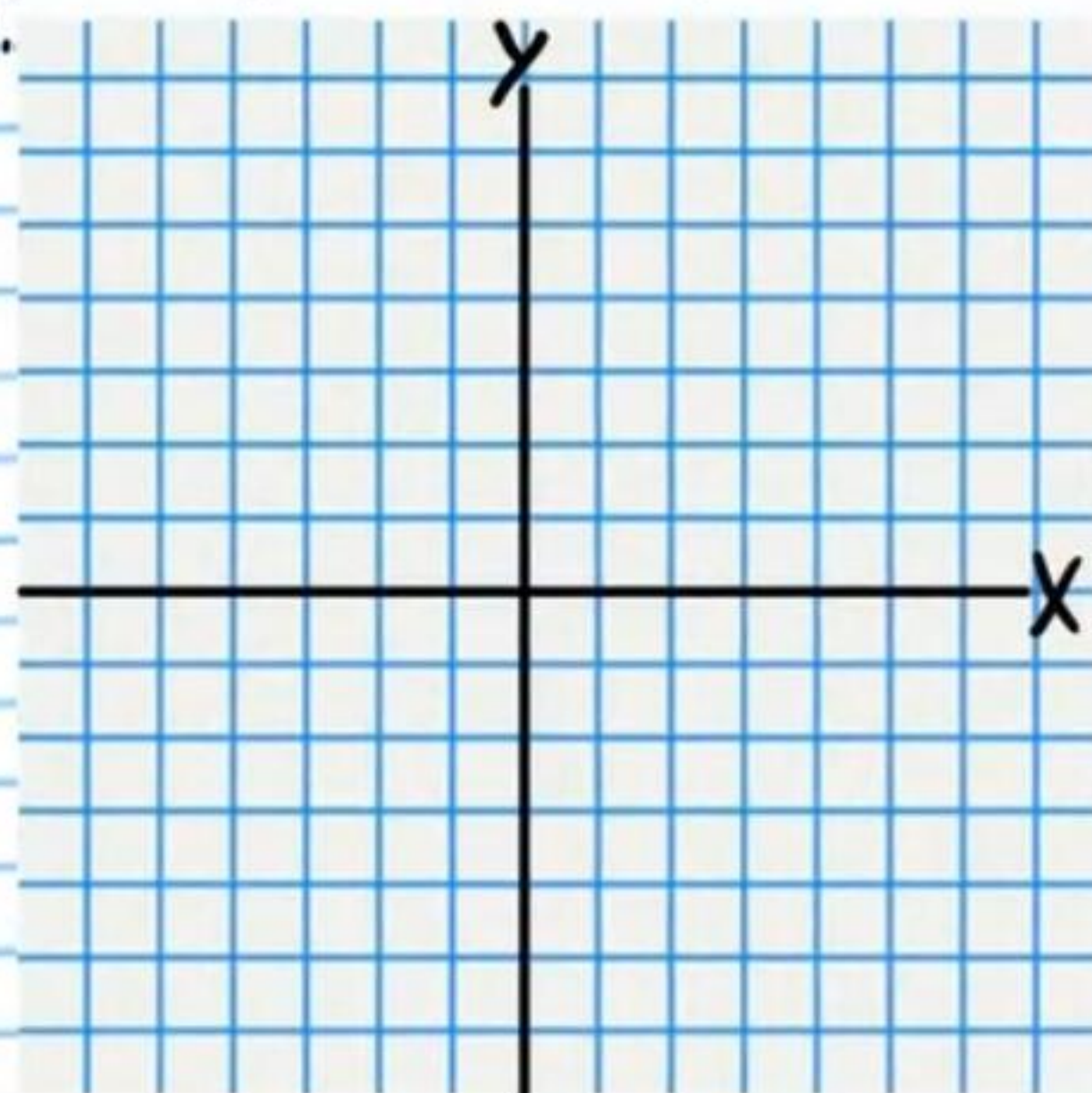


Find equations of circles with the following characteristics and graph each equation.

② Center at $(-2, -2)$ and point on the circle $(0, 0)$.

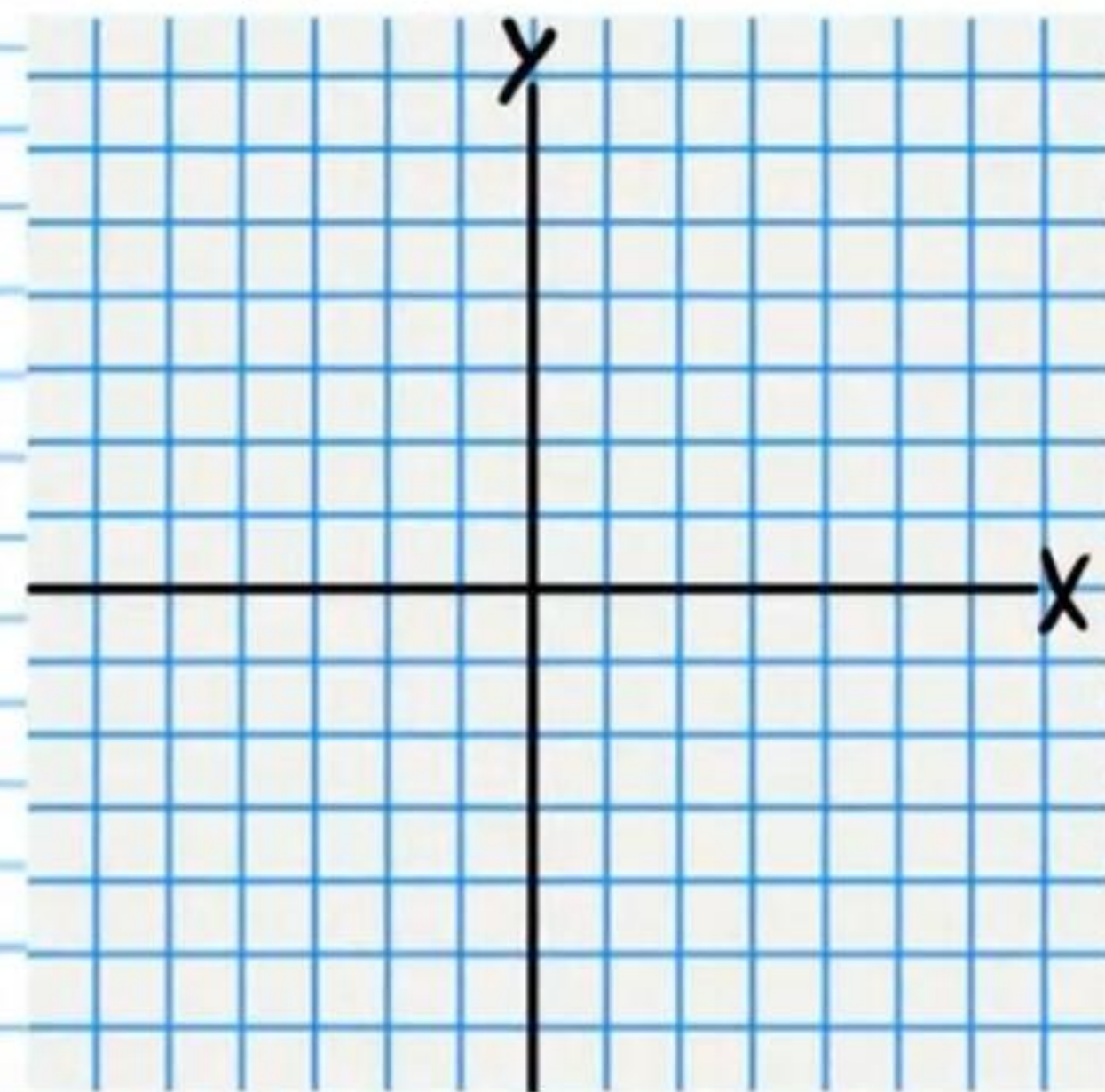


③ Center on the line $y = -1$, radius 3, and point on the circle $(-2, -4)$.

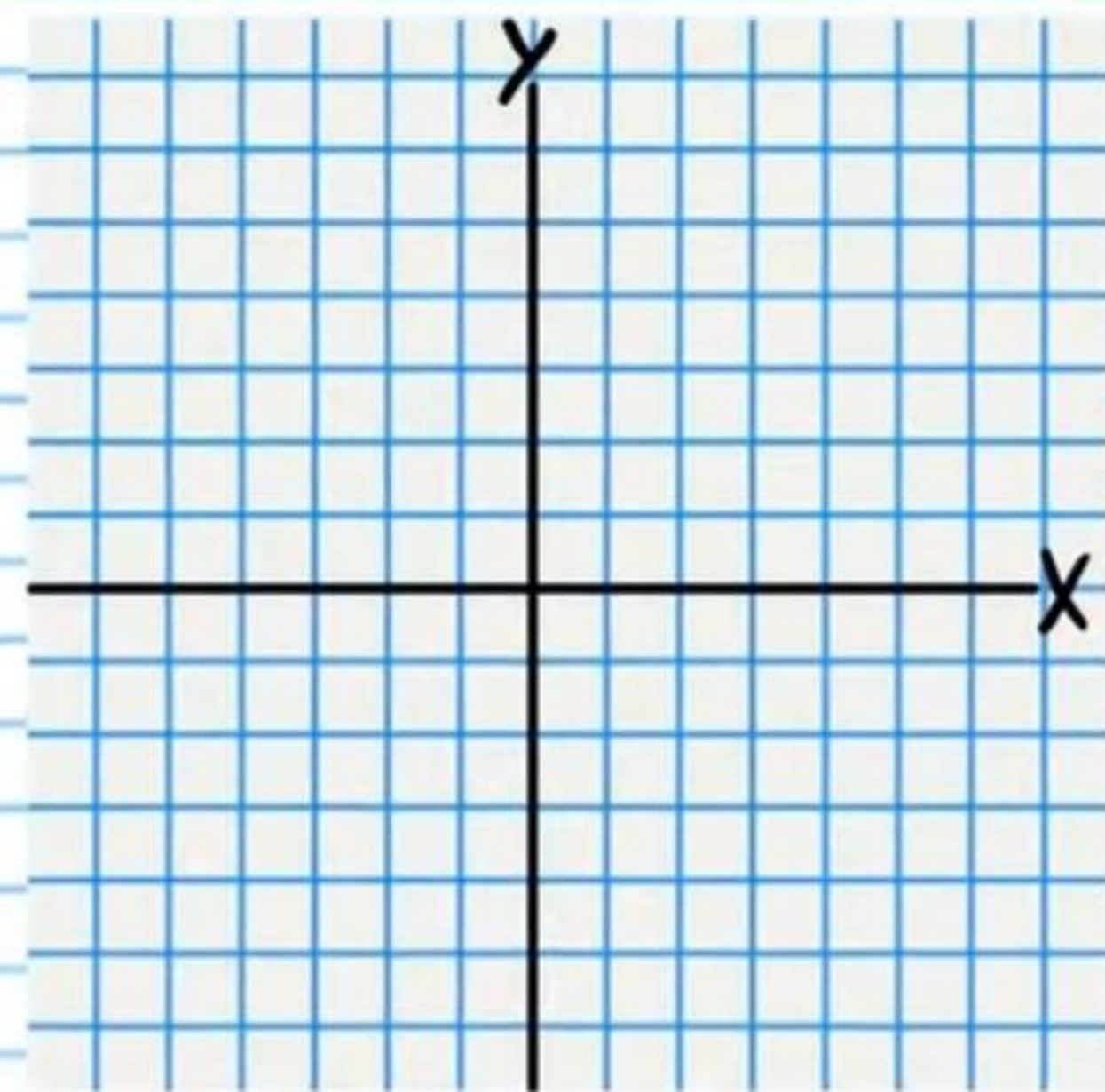


Find equations of parabolas with the following characteristics and graph each equation.

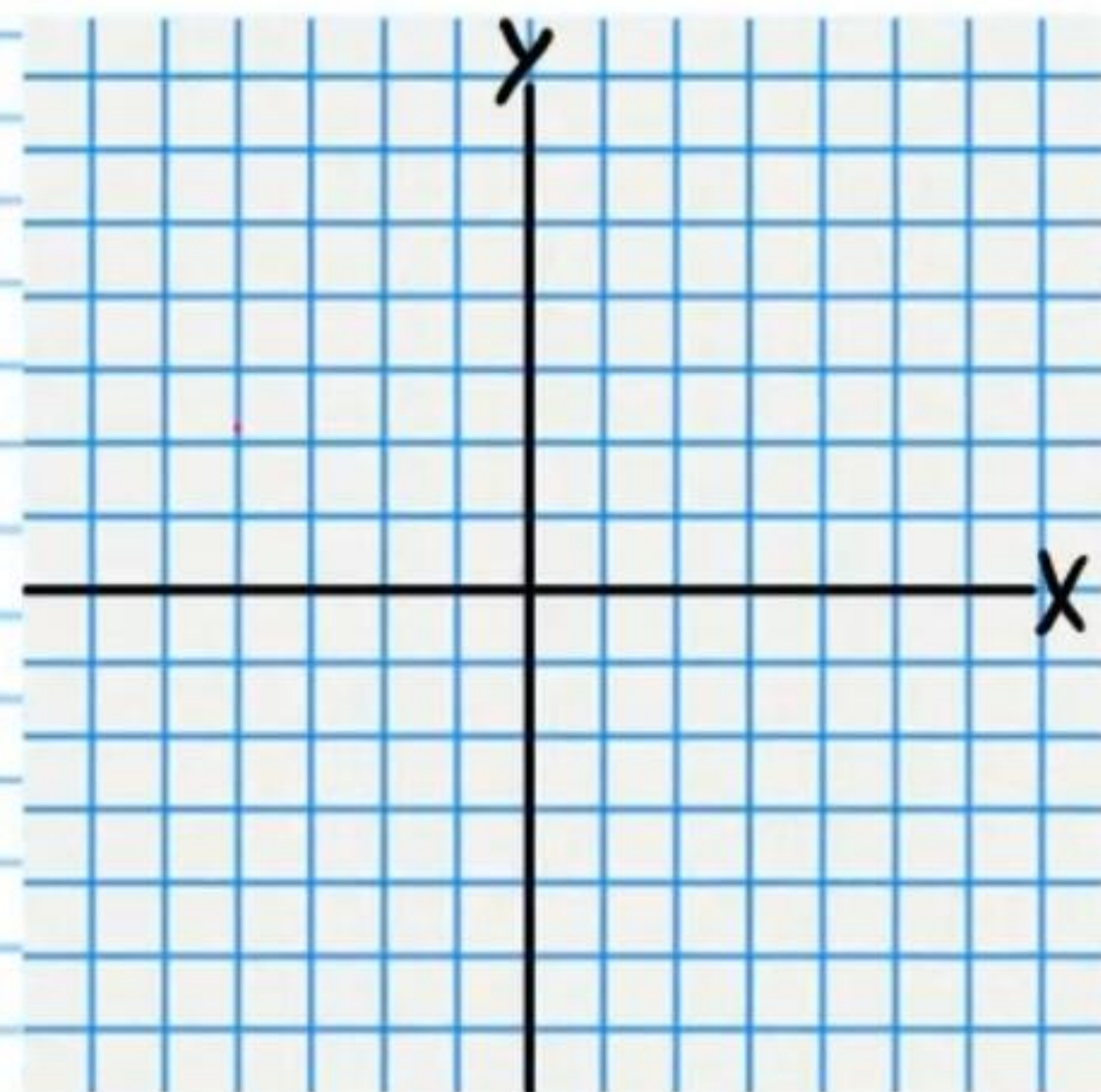
⑪ Vertex $(-3, -4)$ and focus $(-3\frac{1}{3}, -4)$.



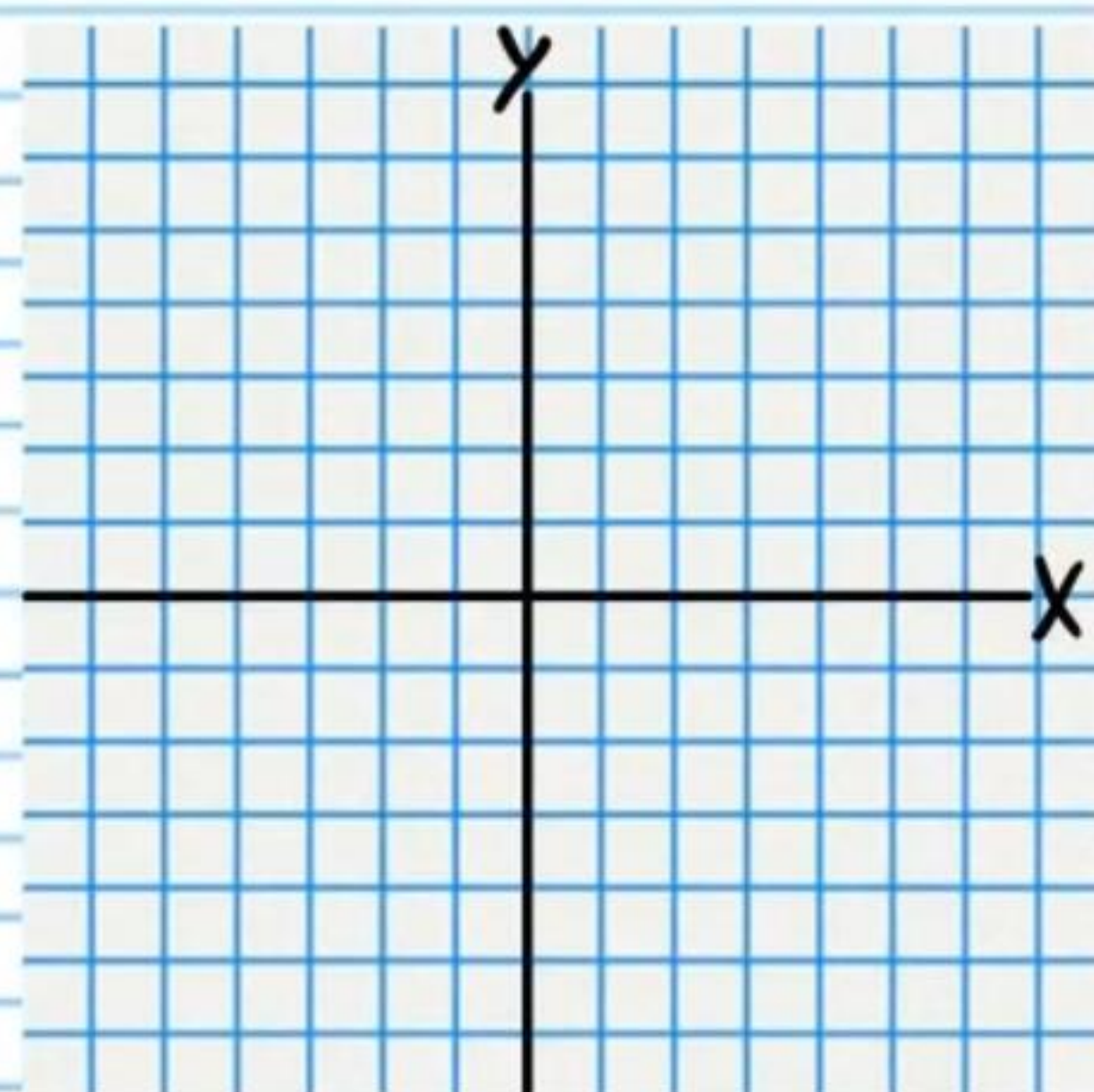
⑫ Vertex $(3, 3)$ and focus $(3, 2)$.



⑬ Vertex $(-4, 2\frac{1}{6})$ and directrix $y = 2$.

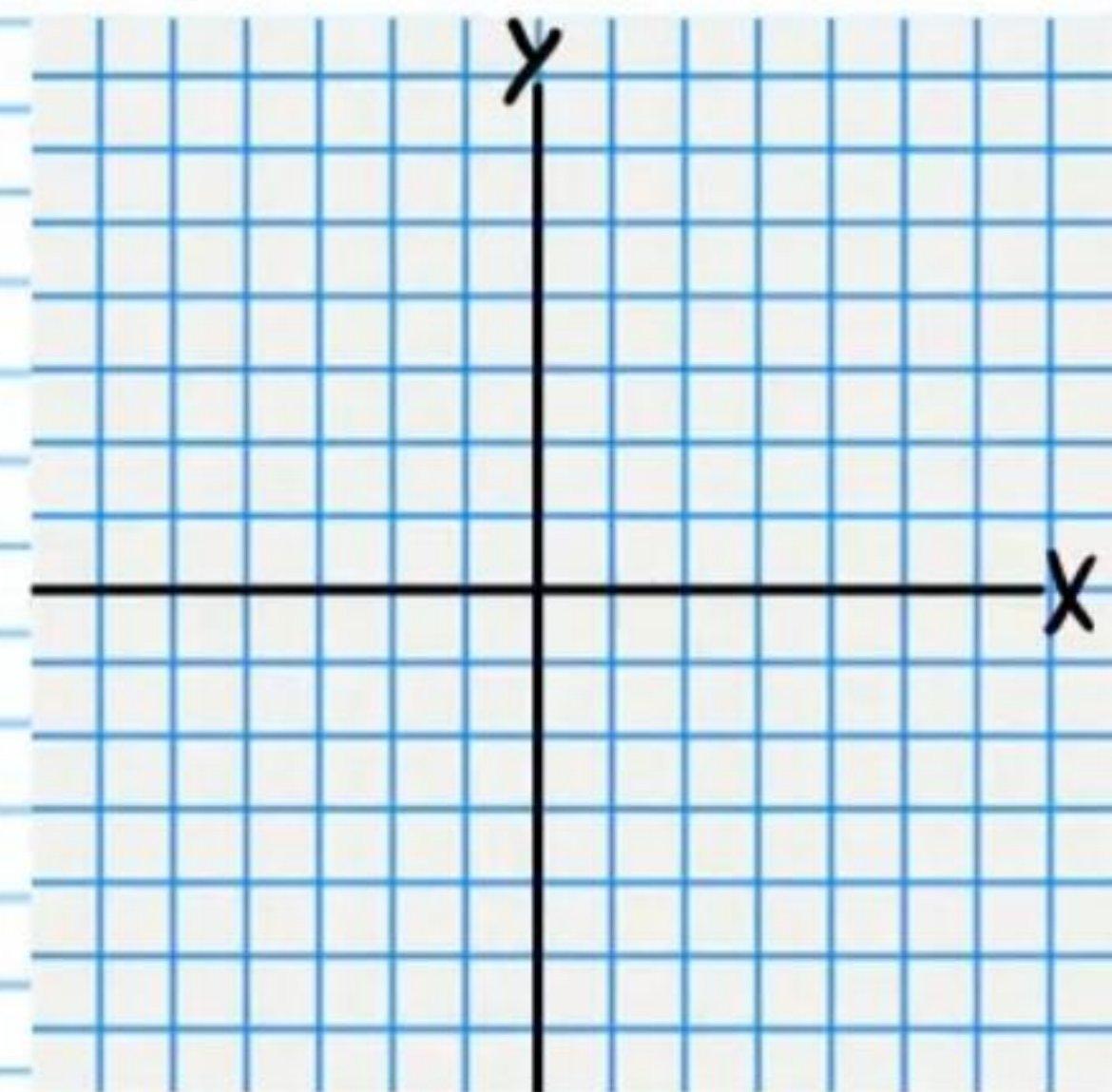


⑭ Focus $(-2, 5)$ and directrix $x = -3$.

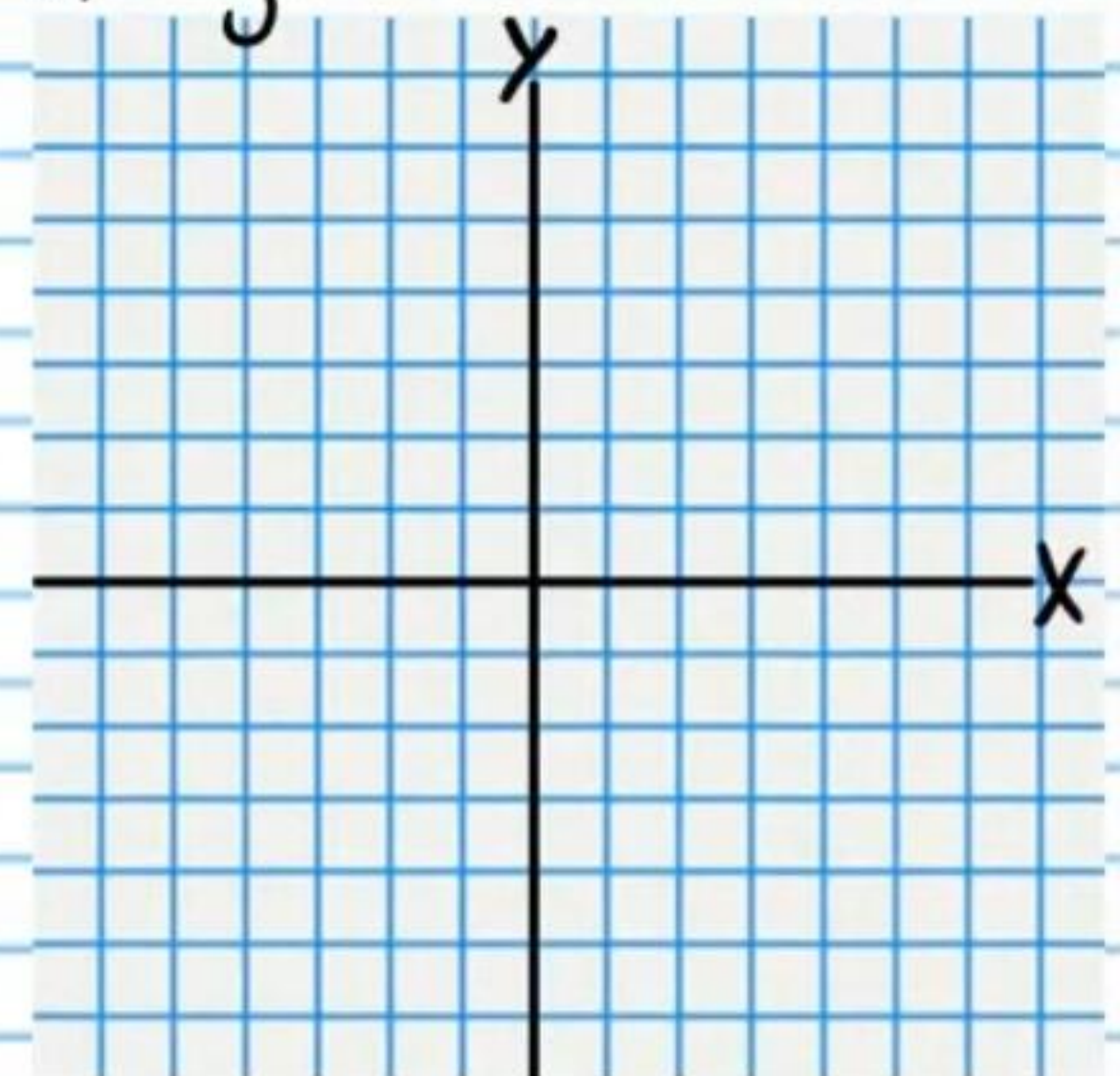


Find equations of ellipses with the following characteristics and graph each equation.

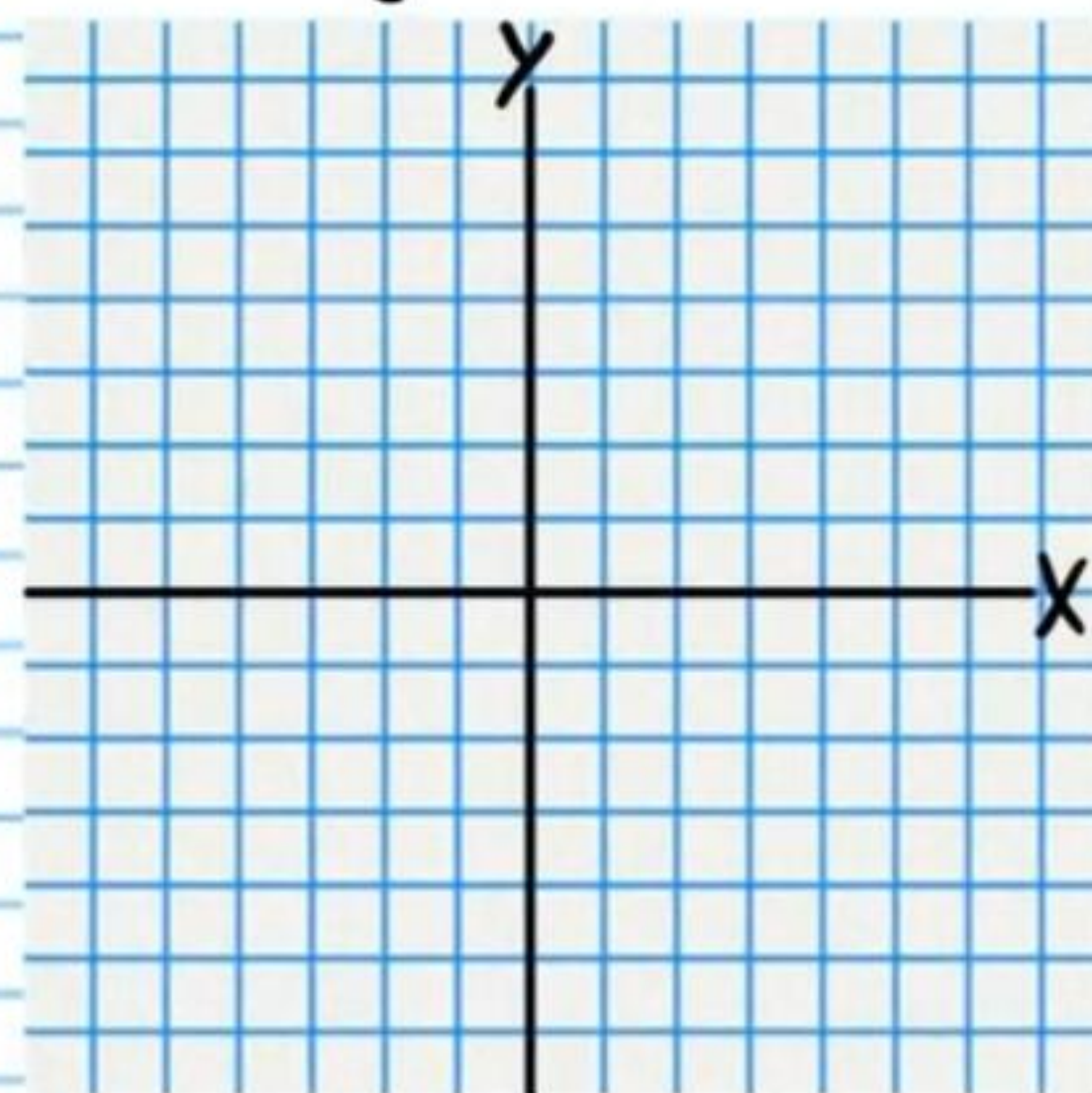
⑮ Foci at $(-5, 3)$ and $(1, 3)$, length of major axis 10.



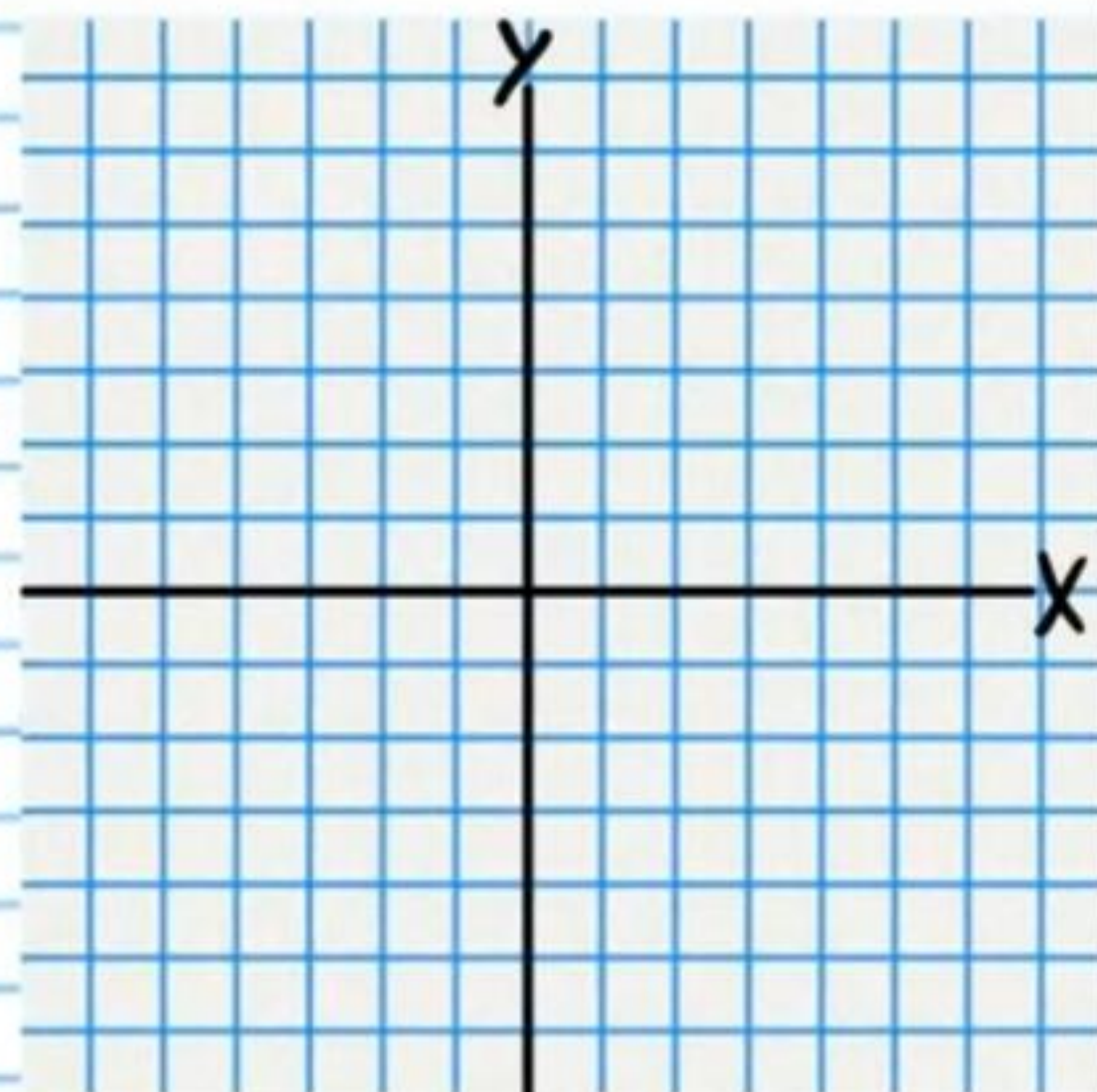
⑯ Vertices at $(4, 2)$ and $(4, -6)$, length of minor axis 6.



⑪ Center $(-2, -3)$, Vertex $(2, -3)$, containing the point $(-2, -2)$.

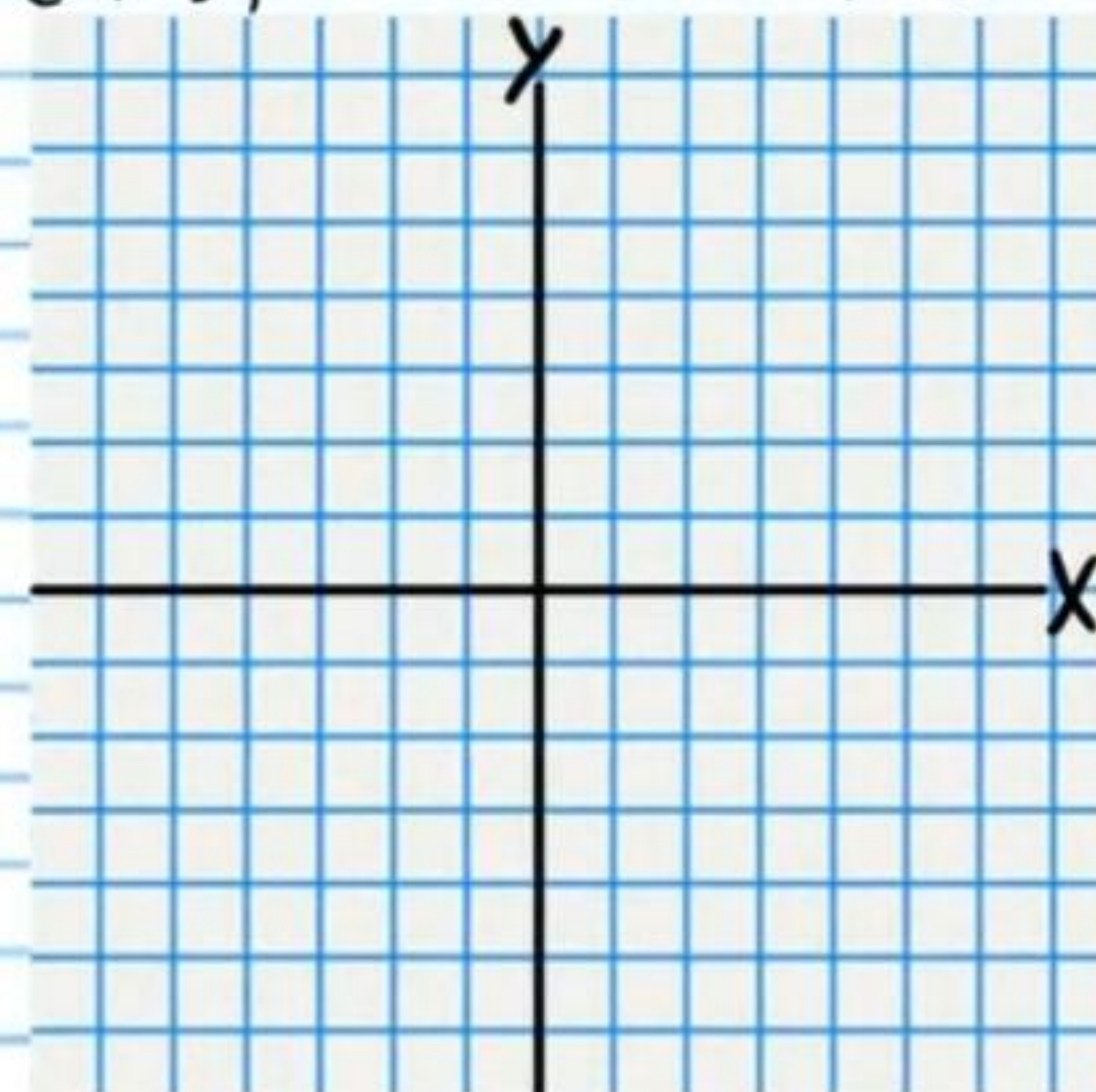


⑫ Vertices $(0, 2)$ and $(0, -2)$, focus $(0, \sqrt{2})$.

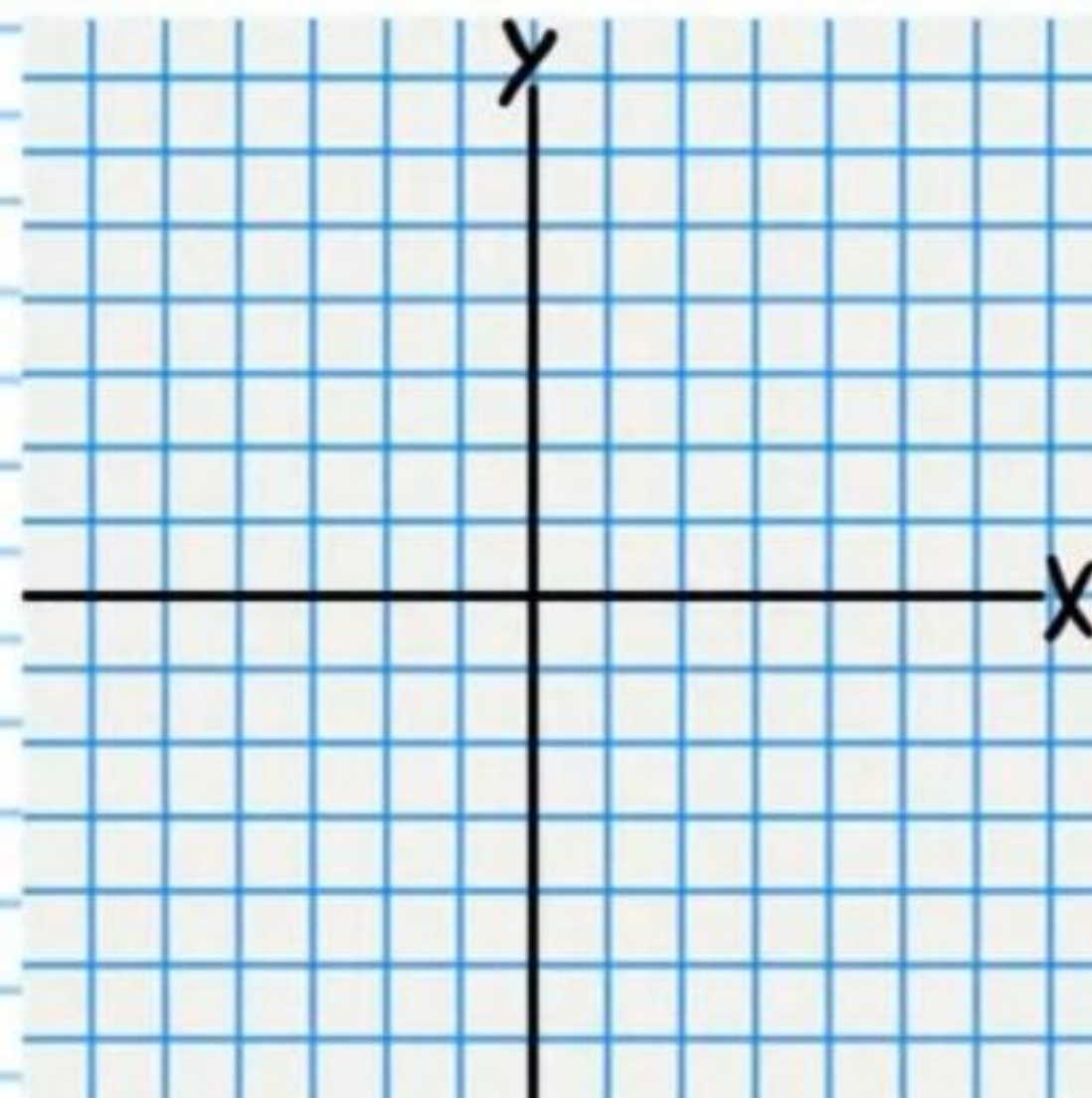


Find equations of hyperbolas with the following characteristics and graph each equation.

⑬ Center at $(3, -3)$, focus at $(3, 0)$, vertex at $(3, -4)$.



⑭ Center at $(-2, -3)$, focus at $(1, -3)$, vertex at $(0, -3)$.



Answers

① $(x-1)^2 + (y+2)^2 = 25$

⑨ $(x+2)^2 + (y+2)^2 = 8$

② $(x+3)^2 + (y+4)^2 = 4$

③ $y-1 = -2(x-5)^2$

④ $x = \frac{1}{8}y^2$

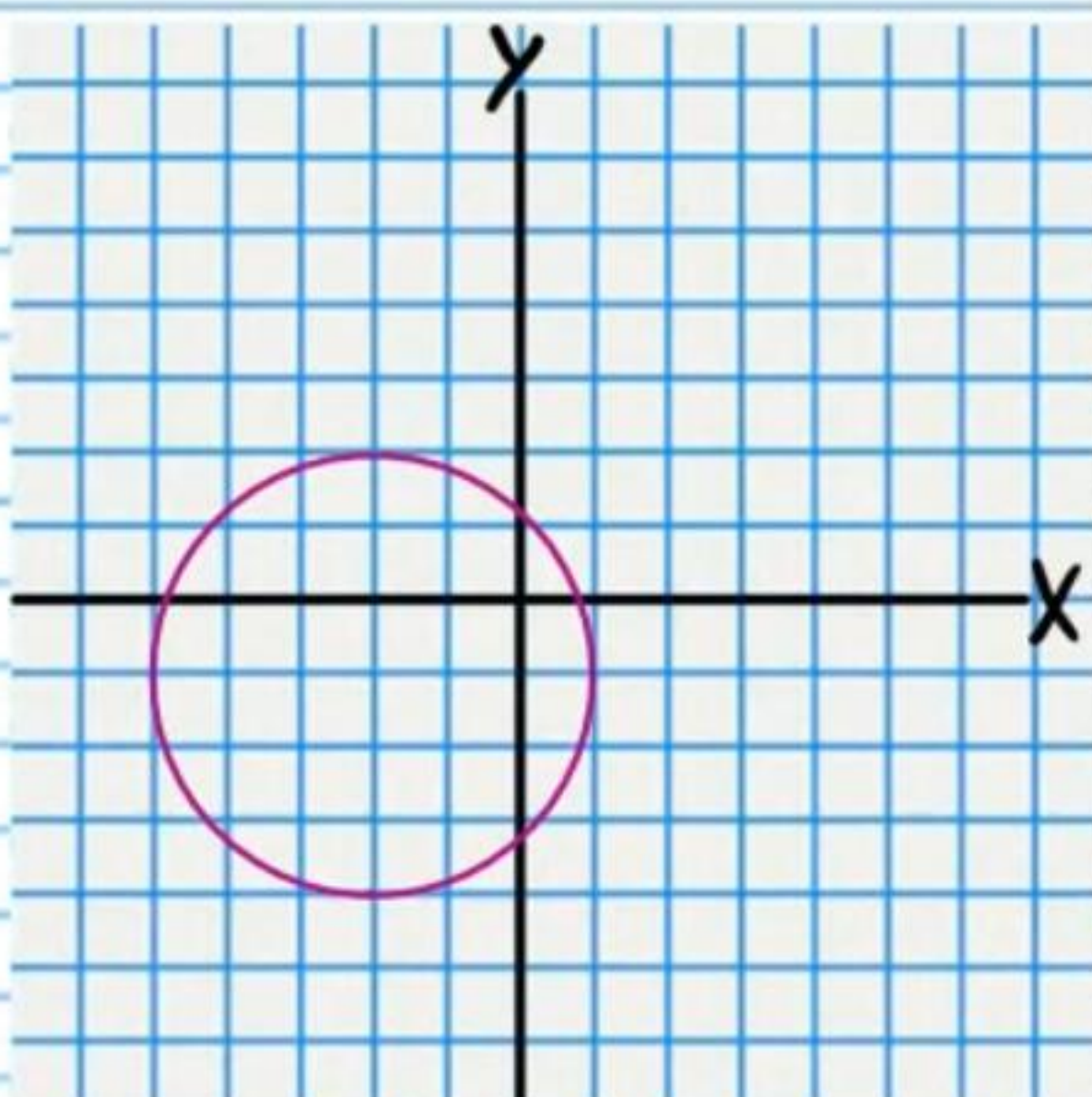
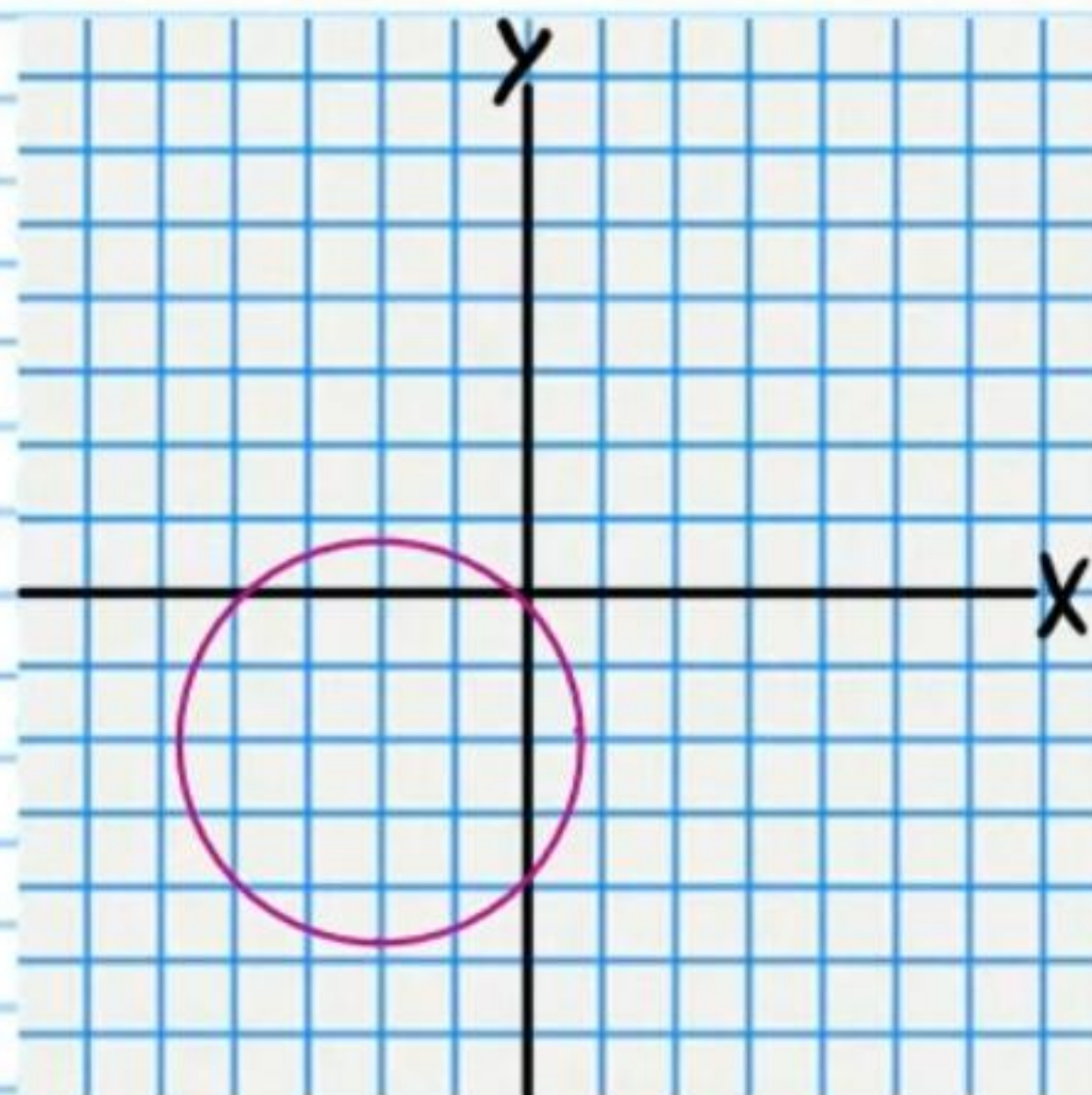
⑤ $\frac{(y-2)^2}{25} + \frac{(x+3)^2}{9} = 1$

⑥ $\frac{(x-1)^2}{36} + (y-2)^2 = 1$

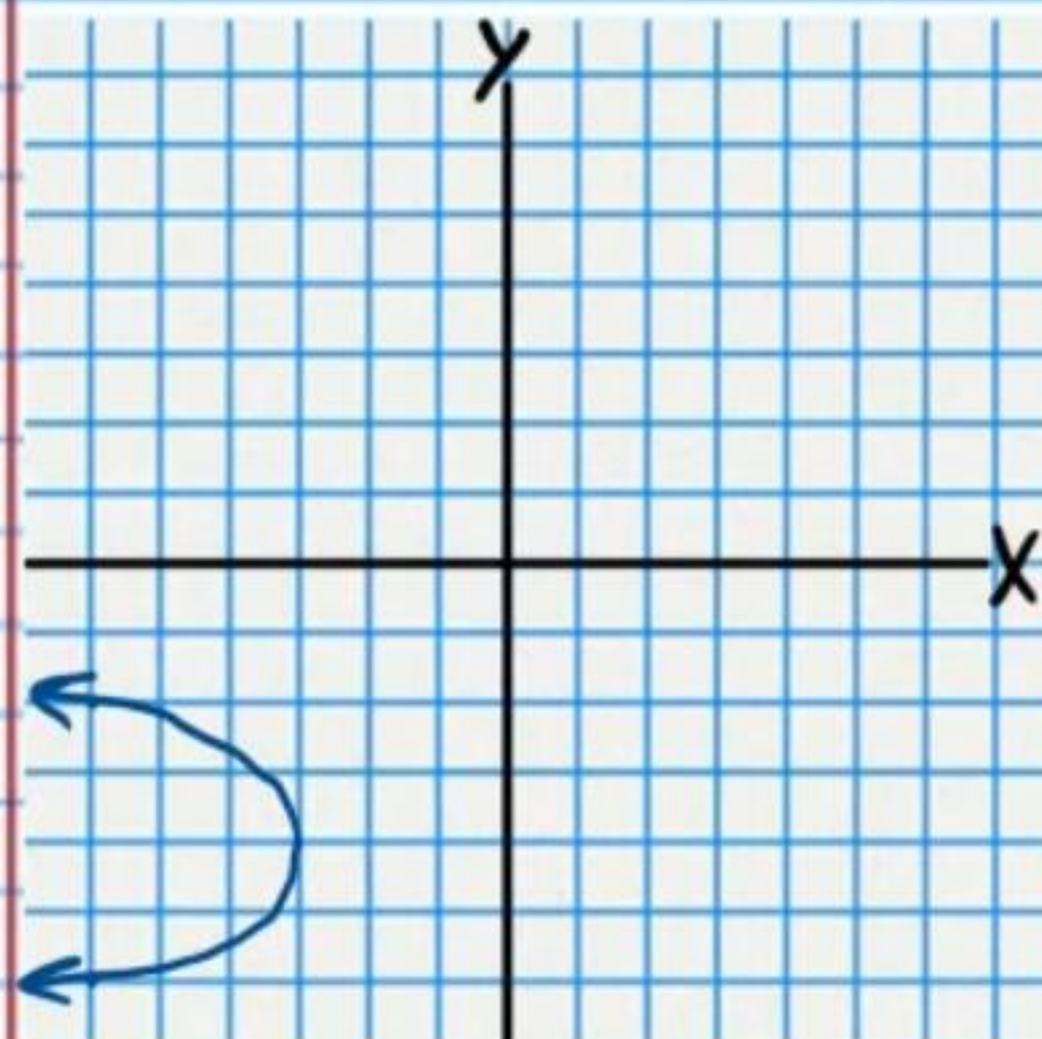
⑦ $\frac{(y-3)^2}{4} - (x+4)^2 = 1$

⑧ $\frac{(x+1)^2}{9} - \frac{(y+2)^2}{9} = 1$

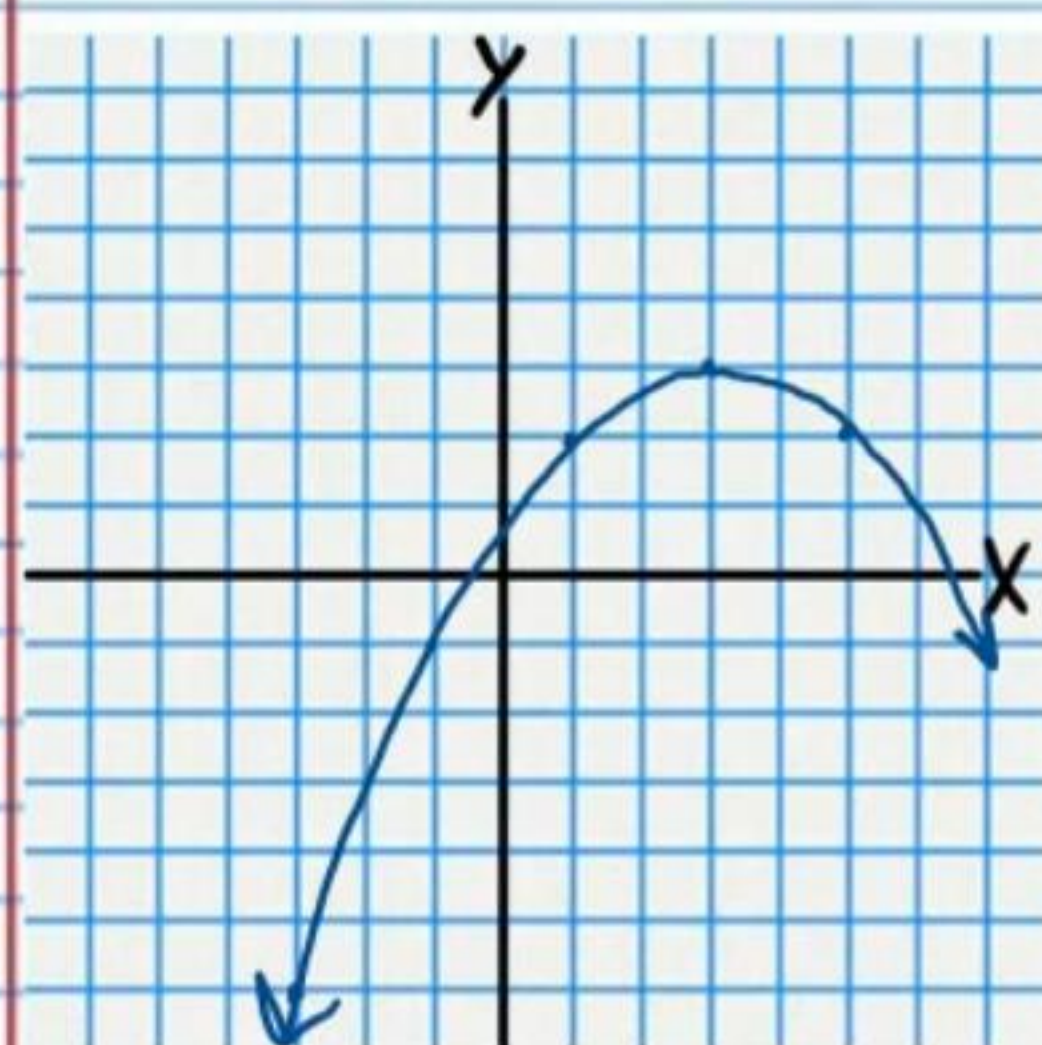
⑩ $(x+2)^2 + (y+1)^2 = 9$



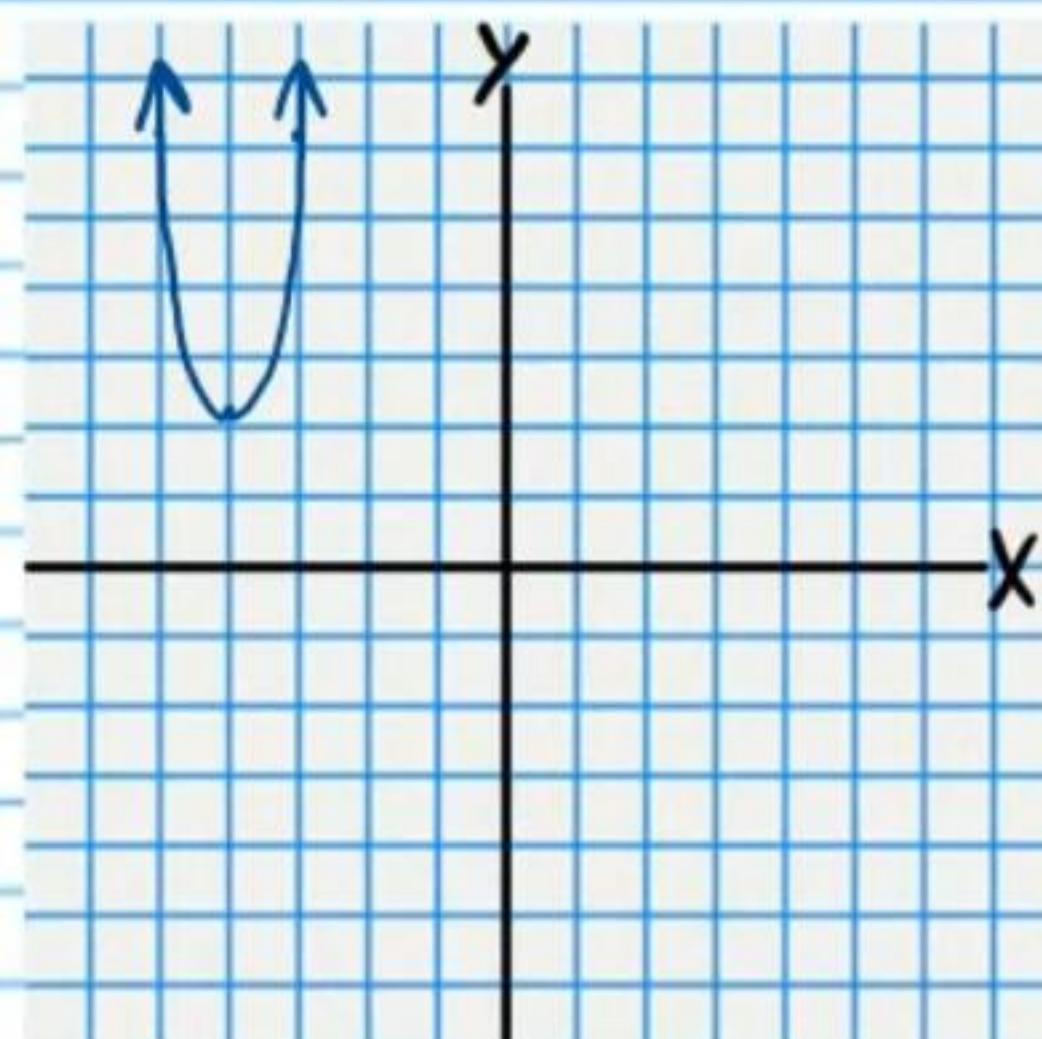
⑪ $x+3 = -\frac{3}{4}(y+4)^2$



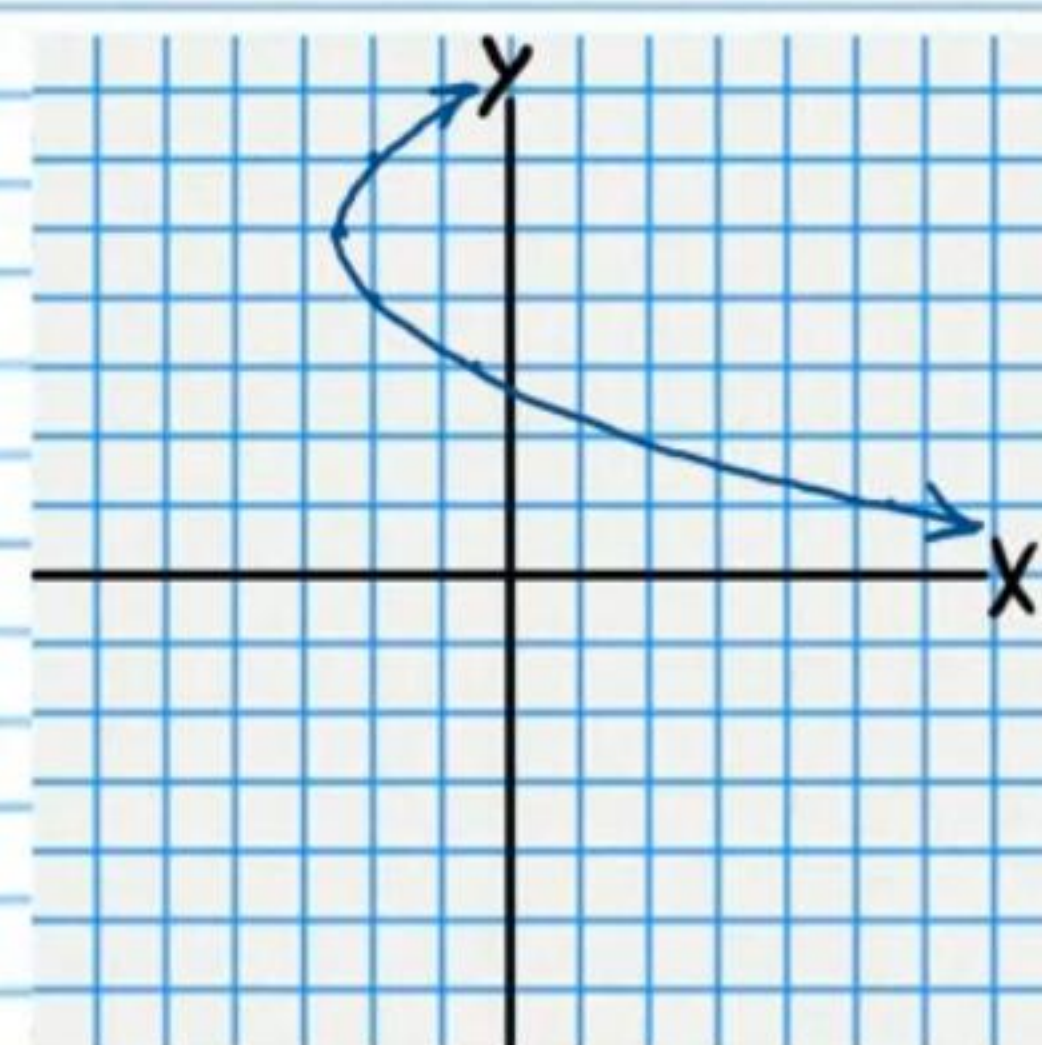
⑫ $y-3 = -\frac{1}{4}(x-3)^2$



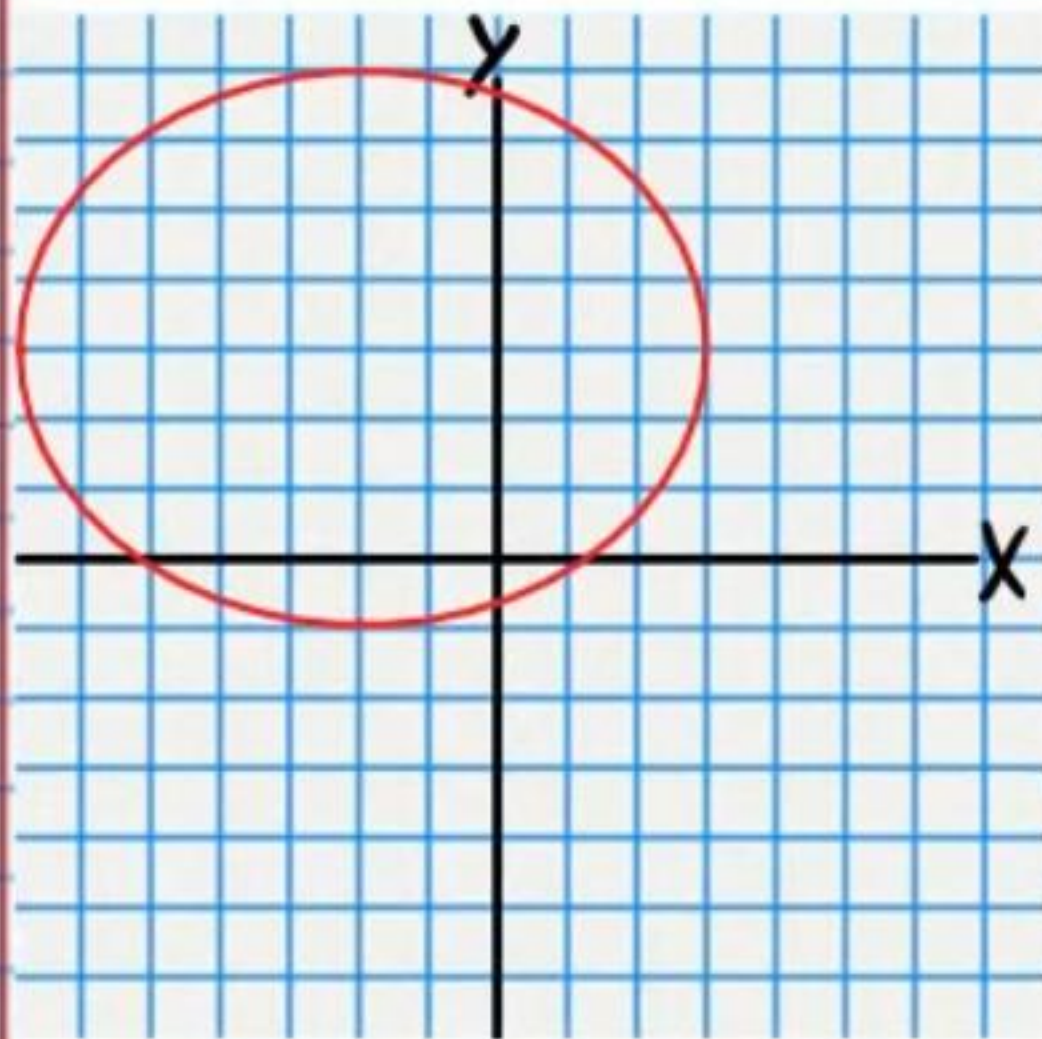
⑬ $y-2\frac{1}{16} = 4(x+4)^2$



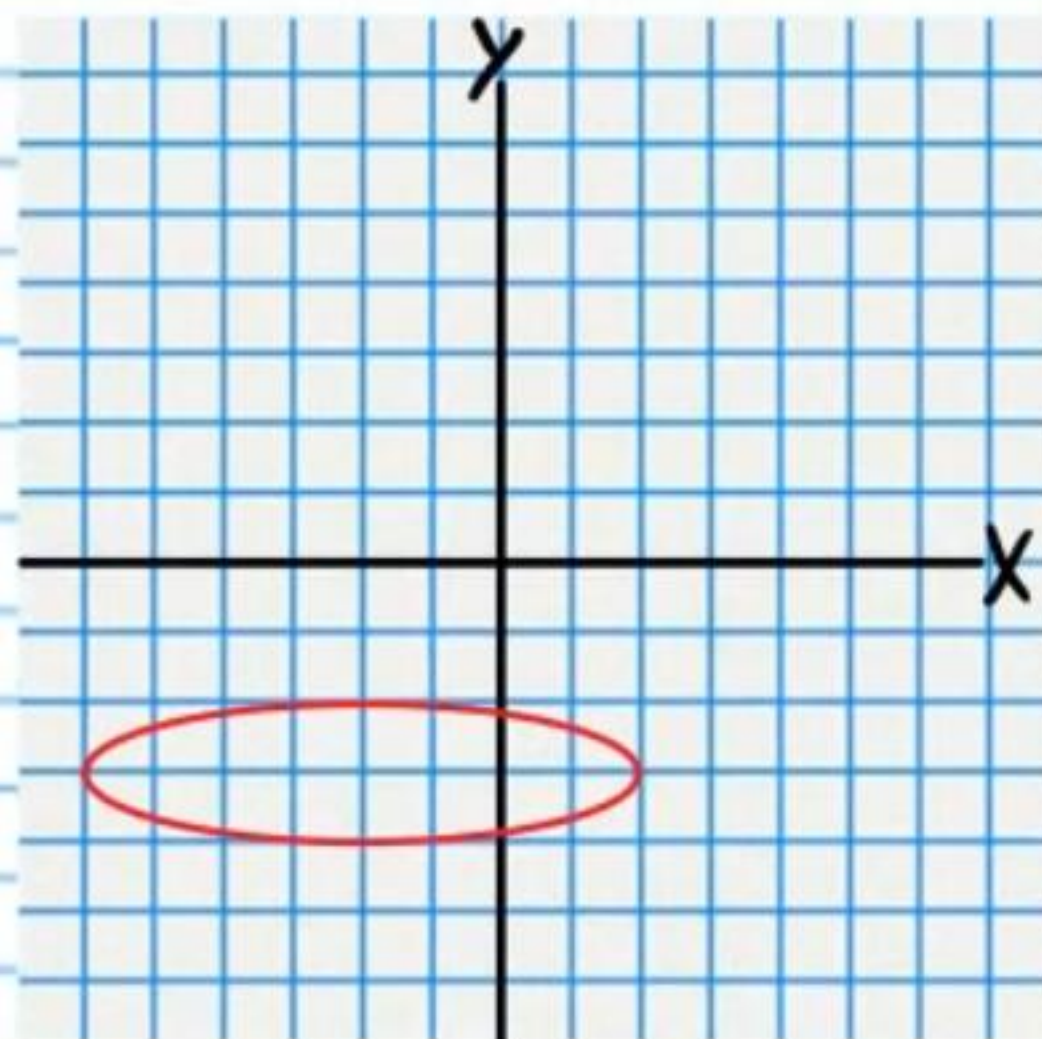
⑭ $x+2\frac{1}{2} = \frac{1}{2}(y-5)^2$



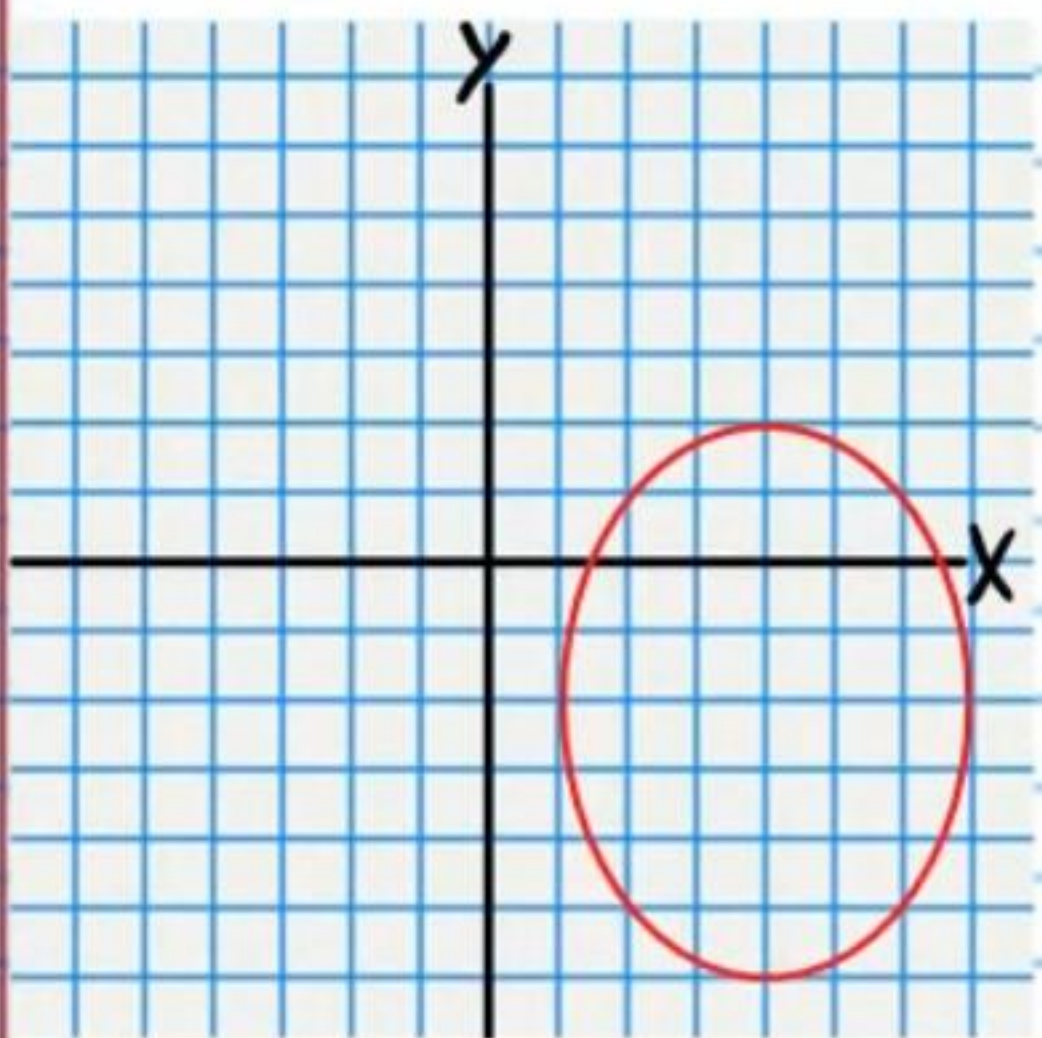
$$(15) \frac{(x+2)^2}{25} + \frac{(y-3)^2}{16} = 1$$



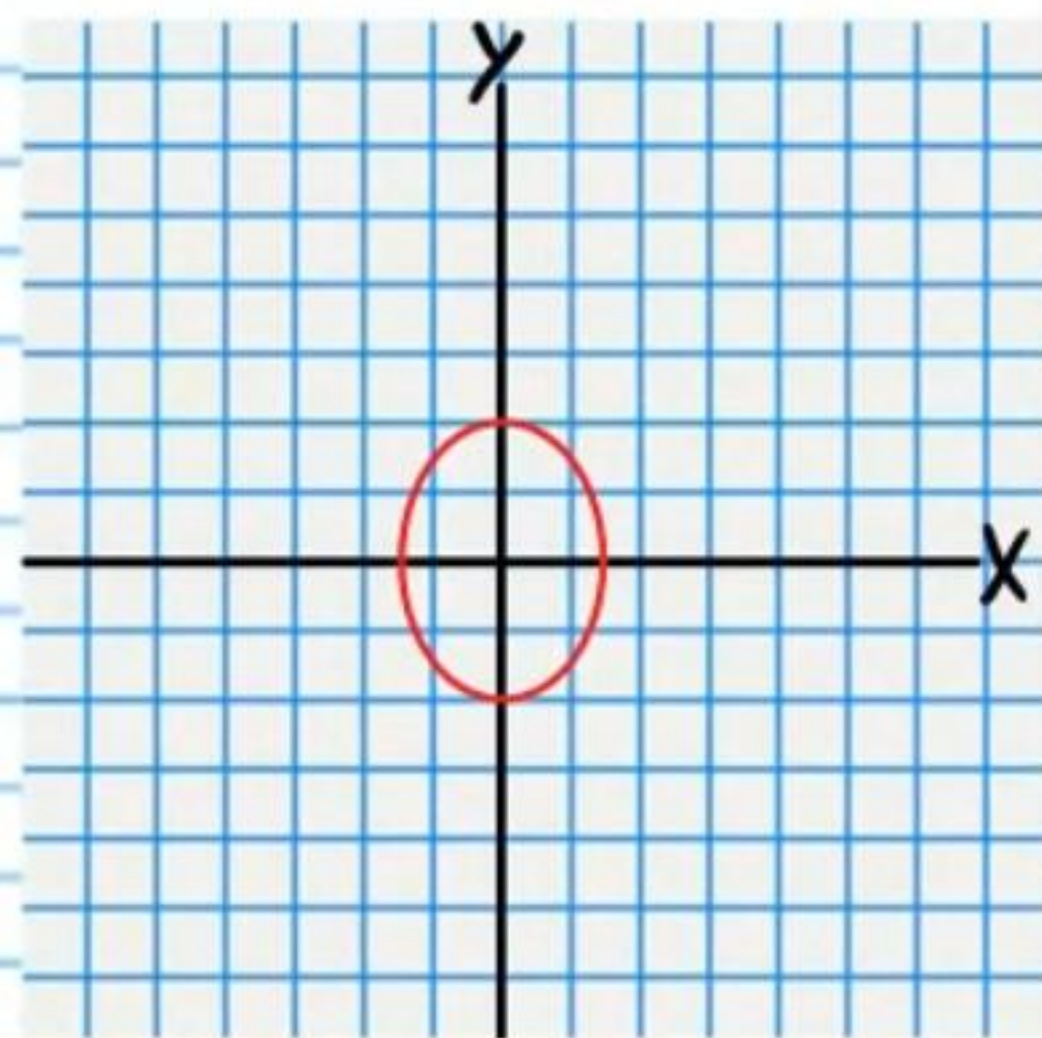
$$(17) \frac{(x+2)^2}{16} + (y+3)^2 = 1$$



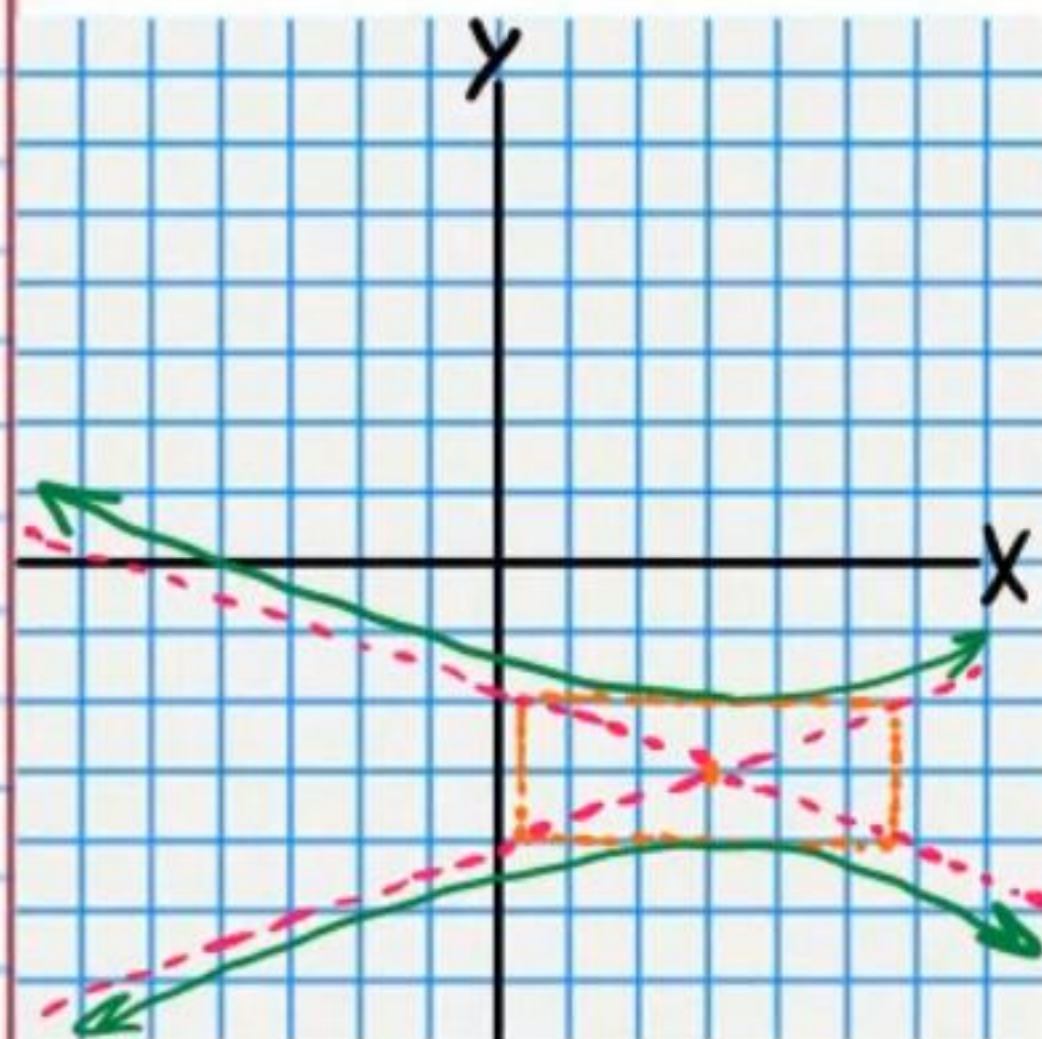
$$(16) \frac{(y+2)^2}{16} + \frac{(x-4)^2}{9} = 1$$



$$(18) \frac{y^2}{4} + \frac{x^2}{2} = 1$$



$$(19) (y+3)^2 - \frac{(x-3)^2}{8} = 1$$



$$(20) \frac{(x+2)^2}{4} - \frac{(y+3)^2}{5} = 1$$

